

SPADE

Speech Across Dialects of English

Structured variability across acoustic dimensions to the intervocalic non-coronal stop voicing contrast in North American and British Isles Englishes

James Tanner, Morgan Sonderegger, Jane Stuart-Smith, Jeff Mielke, Tyler Kendall

BAAP Conference, 8-10th April



University
of Glasgow



McGill

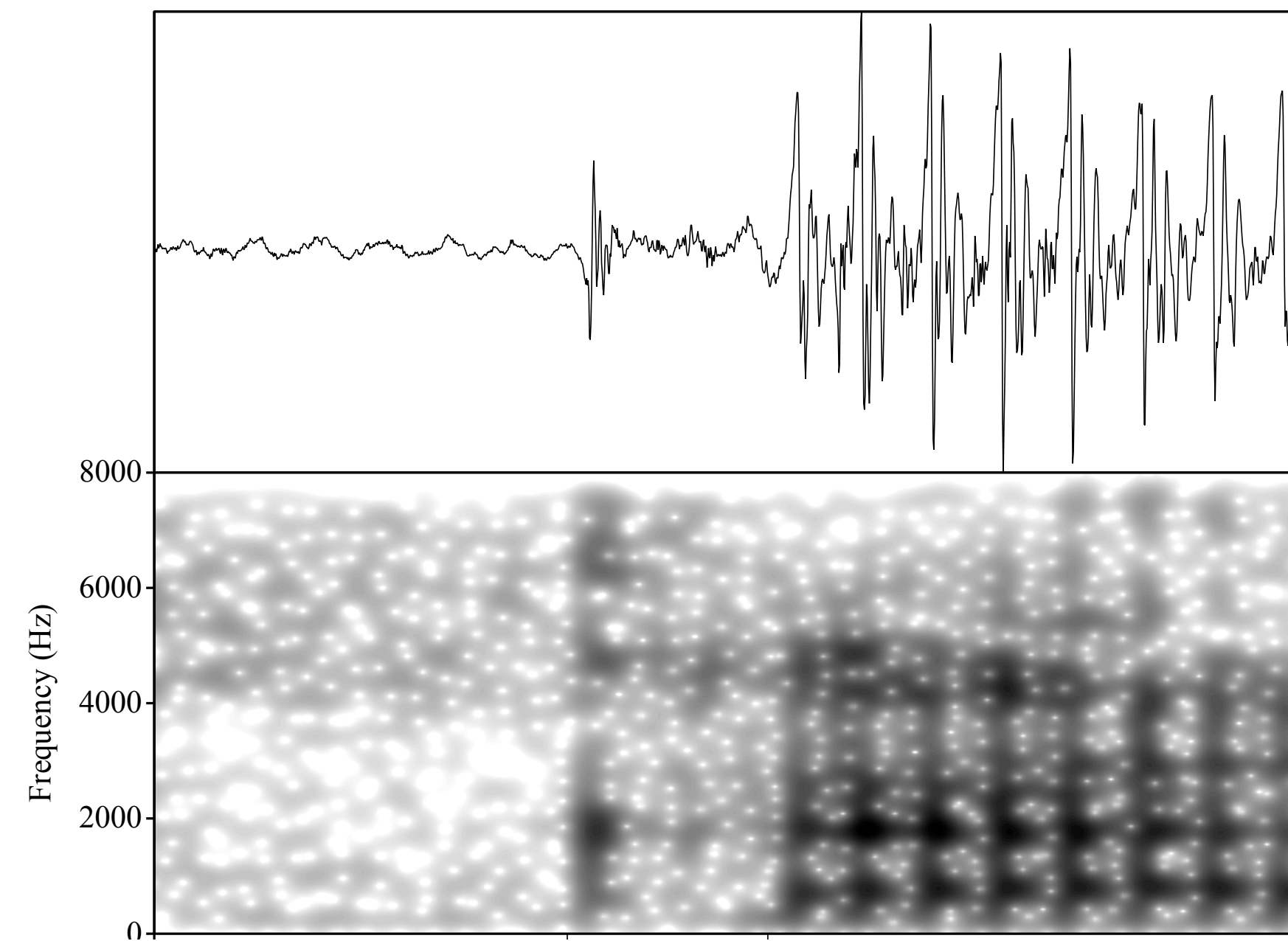
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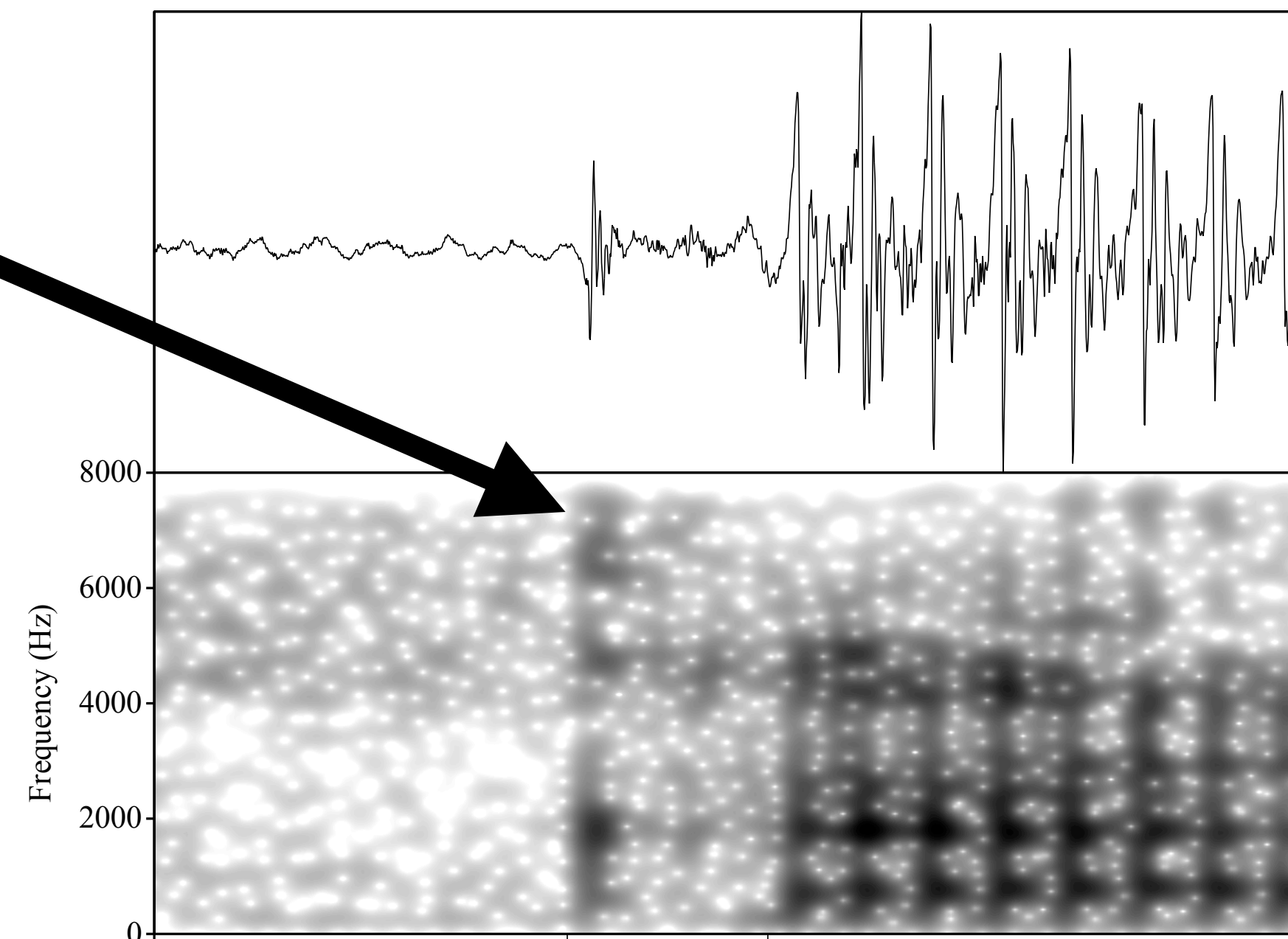
Introduction

- Intervocalic stop voicing (e.g. *rapid~rabid*) signalled by wide range of features
 - Burst presence
 - Duration
 - Closure voicing
 - Spectral intensity
 - F1
 - F0
 - Preceding vowel ratio
 - Following vowel ratio
 - ...



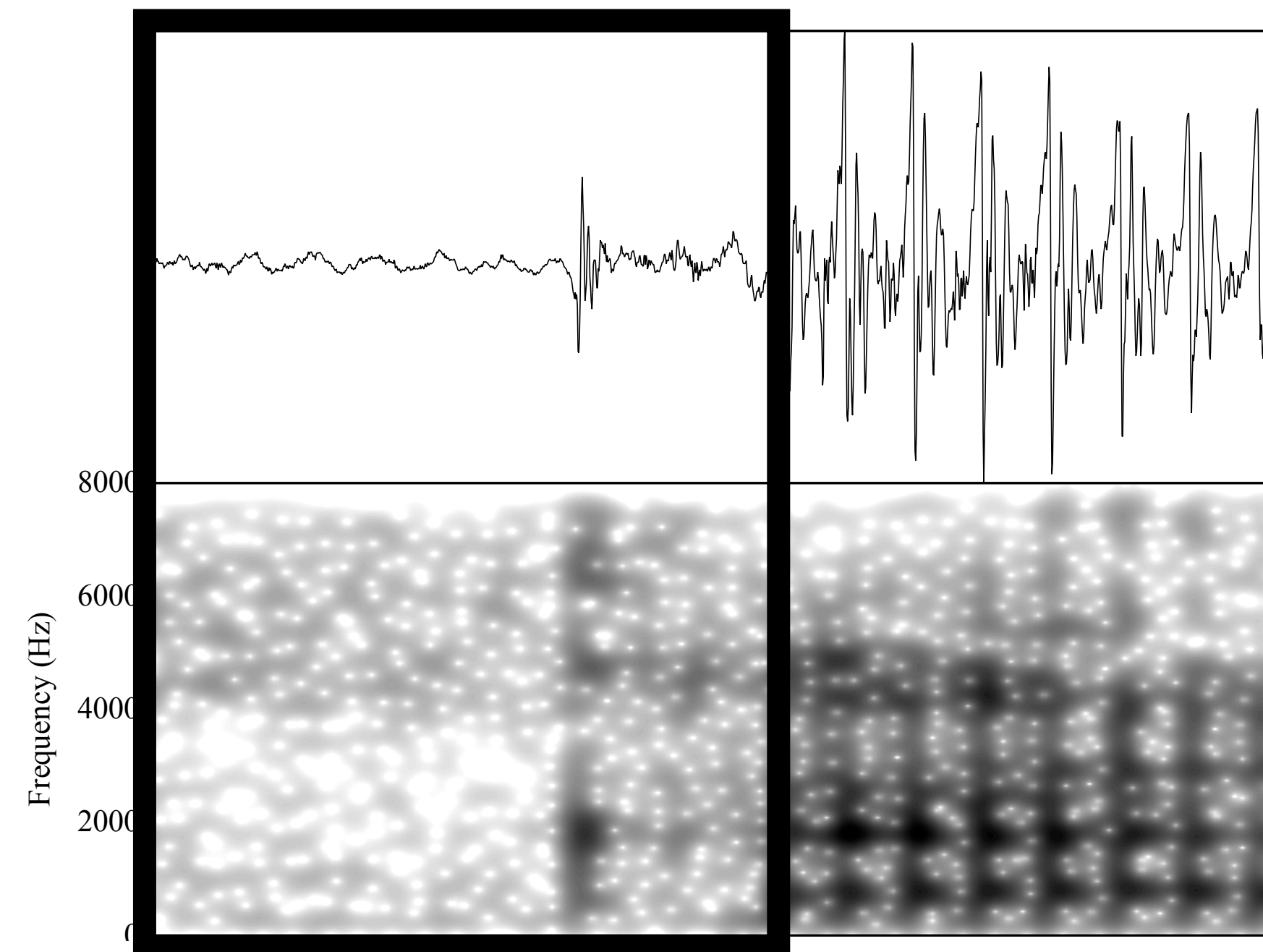
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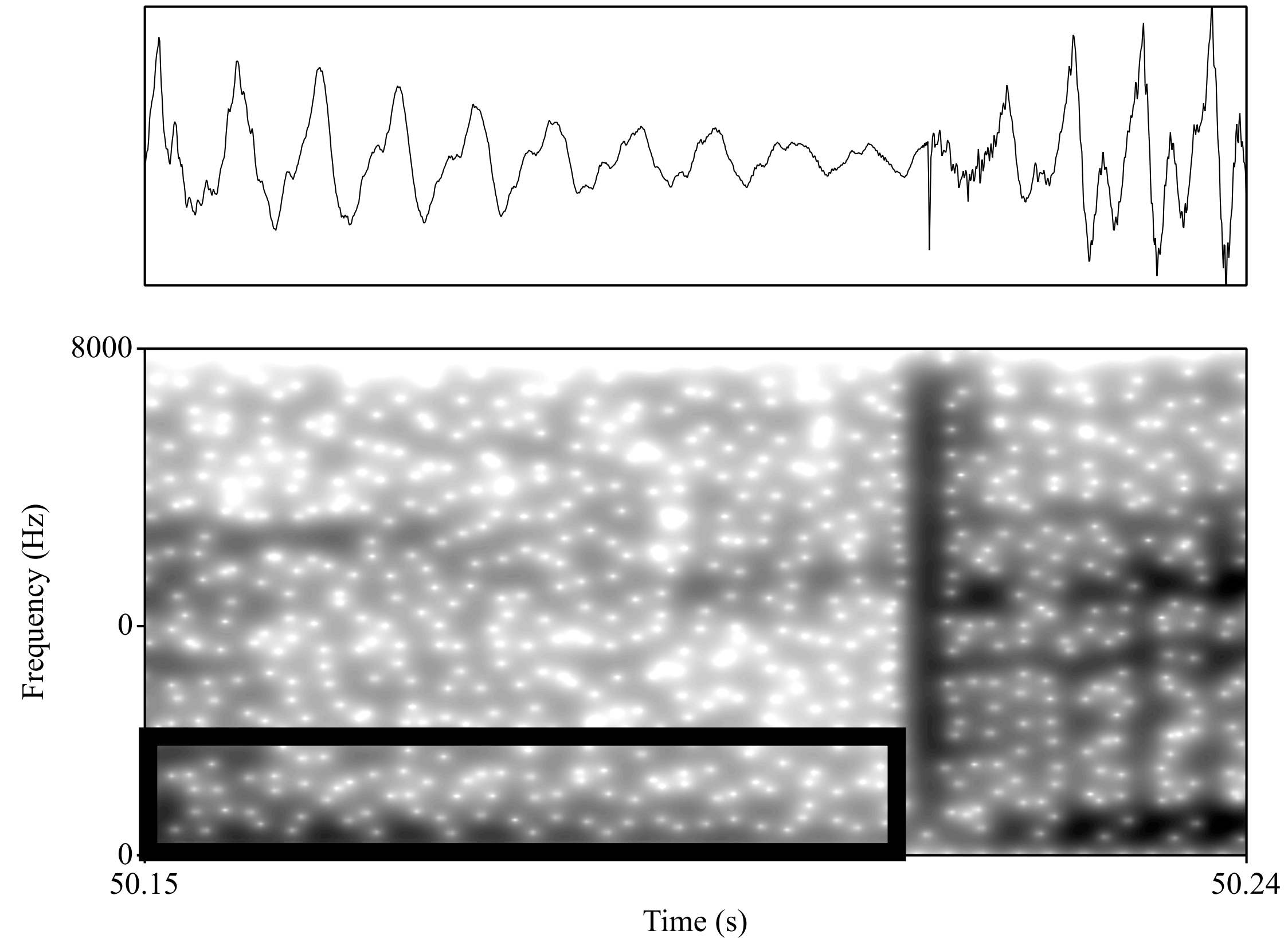
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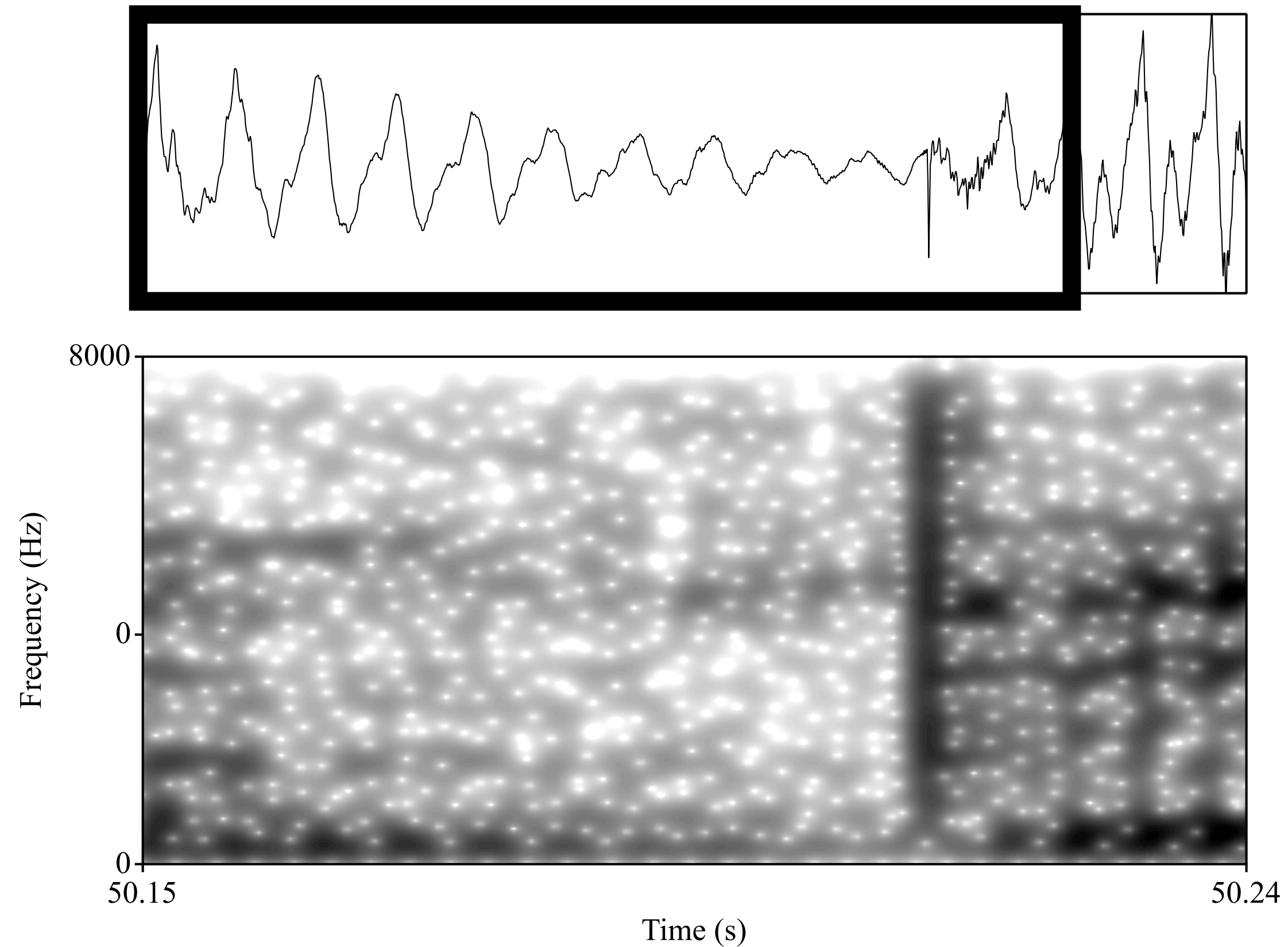
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**How do these vary
across English?**

**How are they related
to each other?**

Introduction

- In English, much interest has focused on socially-salient variation of alveolar stops
- Glottalisation, flapping, coronal stop deletion, etc

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- In English, much interest has focused on socially-salient variation of alveolar stops
 - Glottalisation, flapping, coronal stop deletion, etc
- ➔ **unclear how intervocalic non-coronal voicing contrast (*rapid-rabid*; *backer-bagger*) varies across Englishes**
 - *Do all English dialects realise rapid-rabid similarly?*
 - *Along what dimensions can dialects vary in realising the rapid-rabid contrast?*

Introduction

- **This study:** explore non-coronal stop voicing realisation across large multi-dialectal multi-corpus English speech dataset
- **RQ 1:** *how consistent is the realisation of phonological voicing across English dialects?*
- **RQ 2:** *how are multiple acoustic properties related to each other in signalling the contrast across dialects?*

Data

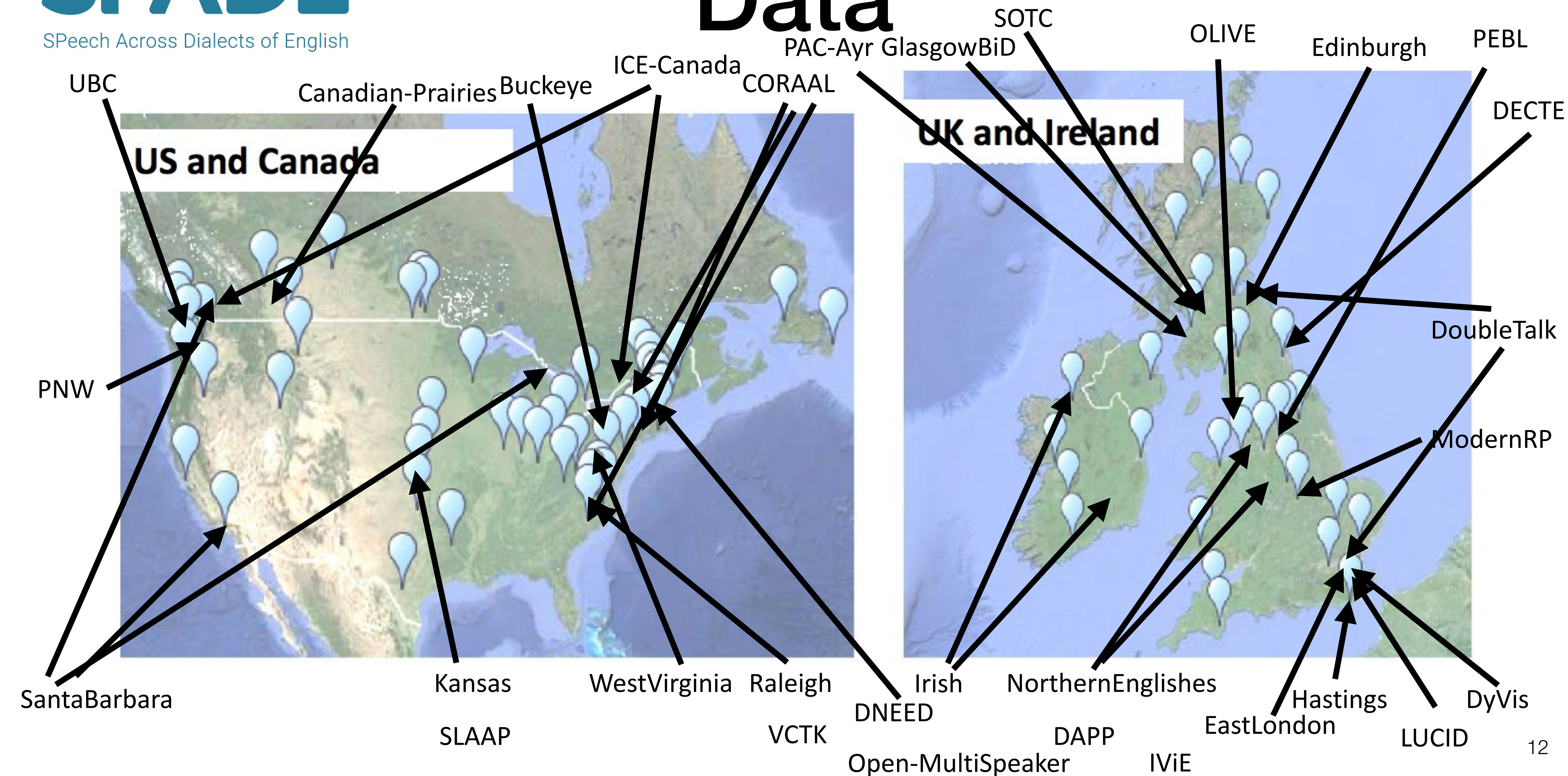
US and Canada



UK and Ireland



Data



Data

- 15 North American and British Isles English dialects from 31 speech corpora (~5400 speakers)
- ~123k intervocalic labial {/p/, /b/} and velar {/k/, /g/} stops
- Unstressed syllable onsets (e.g. *rapid*, *rabid*, *slacker*, etc)

Canada (East)

Canada (West)

England (Midlands)

England (Northern)

England (Southern)

Ireland (North)

Ireland (South)

Scotland (Central West East)

Scotland (Highlands)

Scotland (Northern Insular)

US (Northern)

US (South)

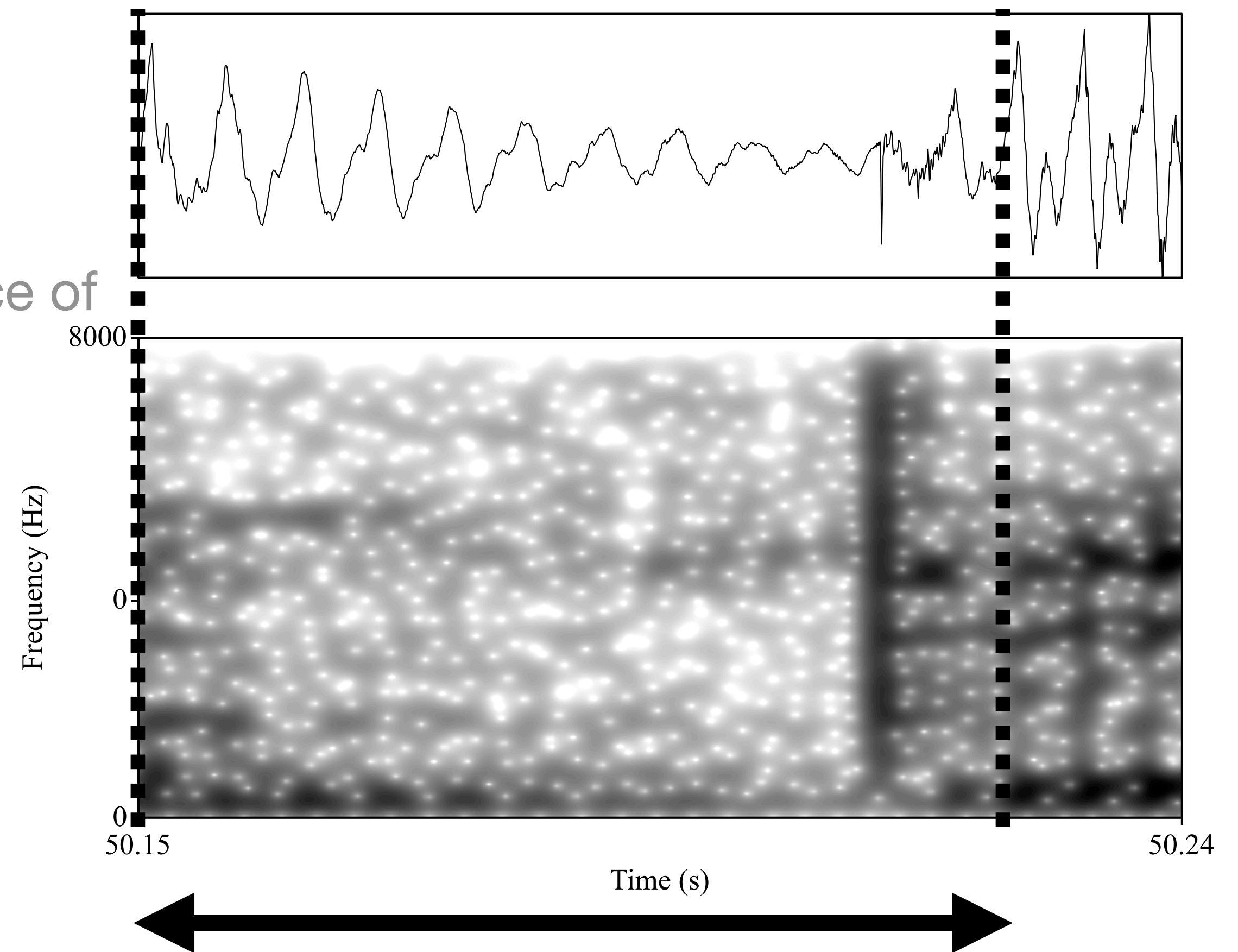
US (West & Midland)

Wales (North)

Wales (South)

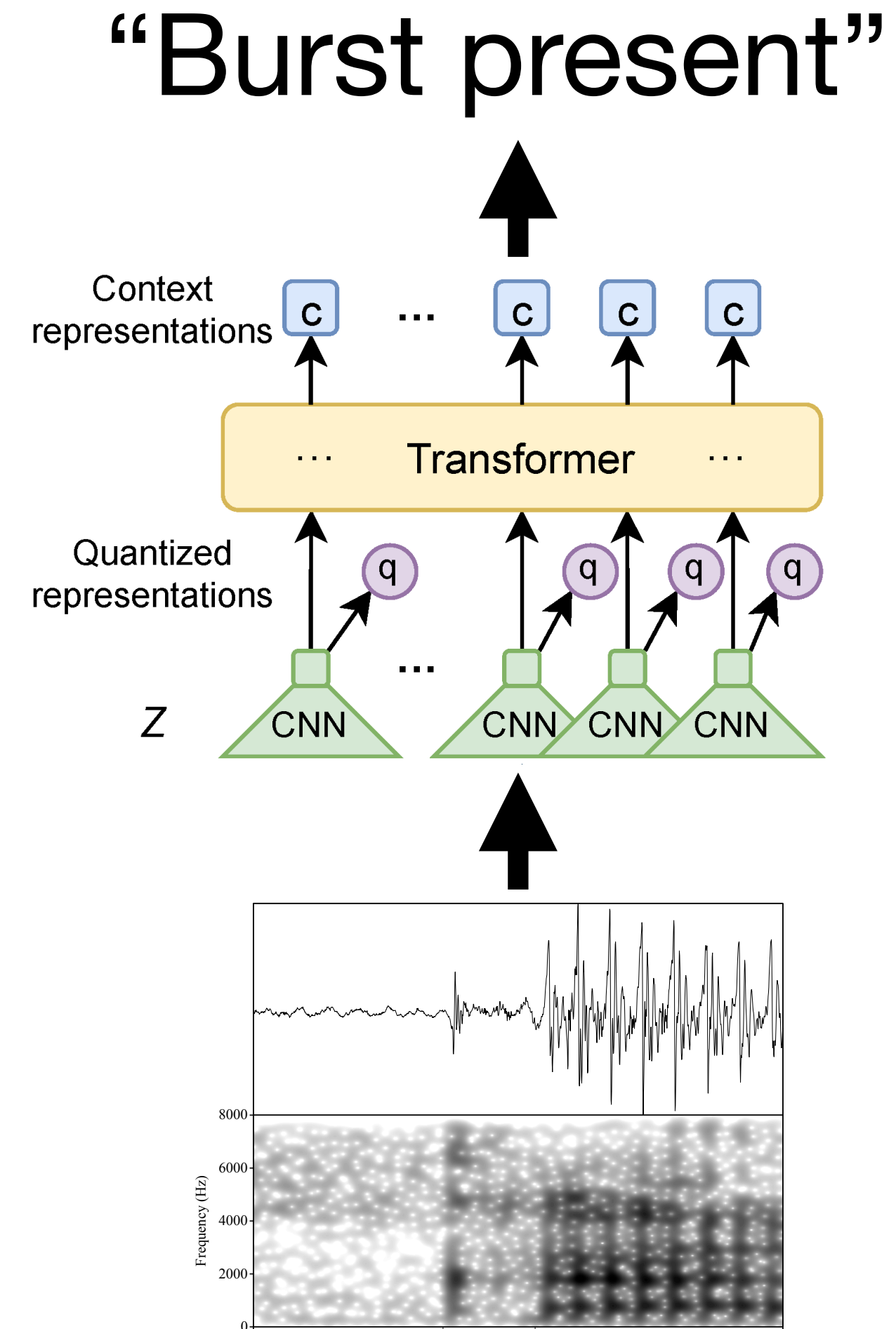
Measures

- Stop segment duration
- Presence of stop burst
 - Estimated from pre-trained wav2vec2.0 model: predicts binary presence/absence of stop burst
- Intensity minimum
 - Reflects acoustic degree of closure
- Voicing During Interval (VDI)
 - Praat VoiceReport 'number of unvoiced frames' across stop interval



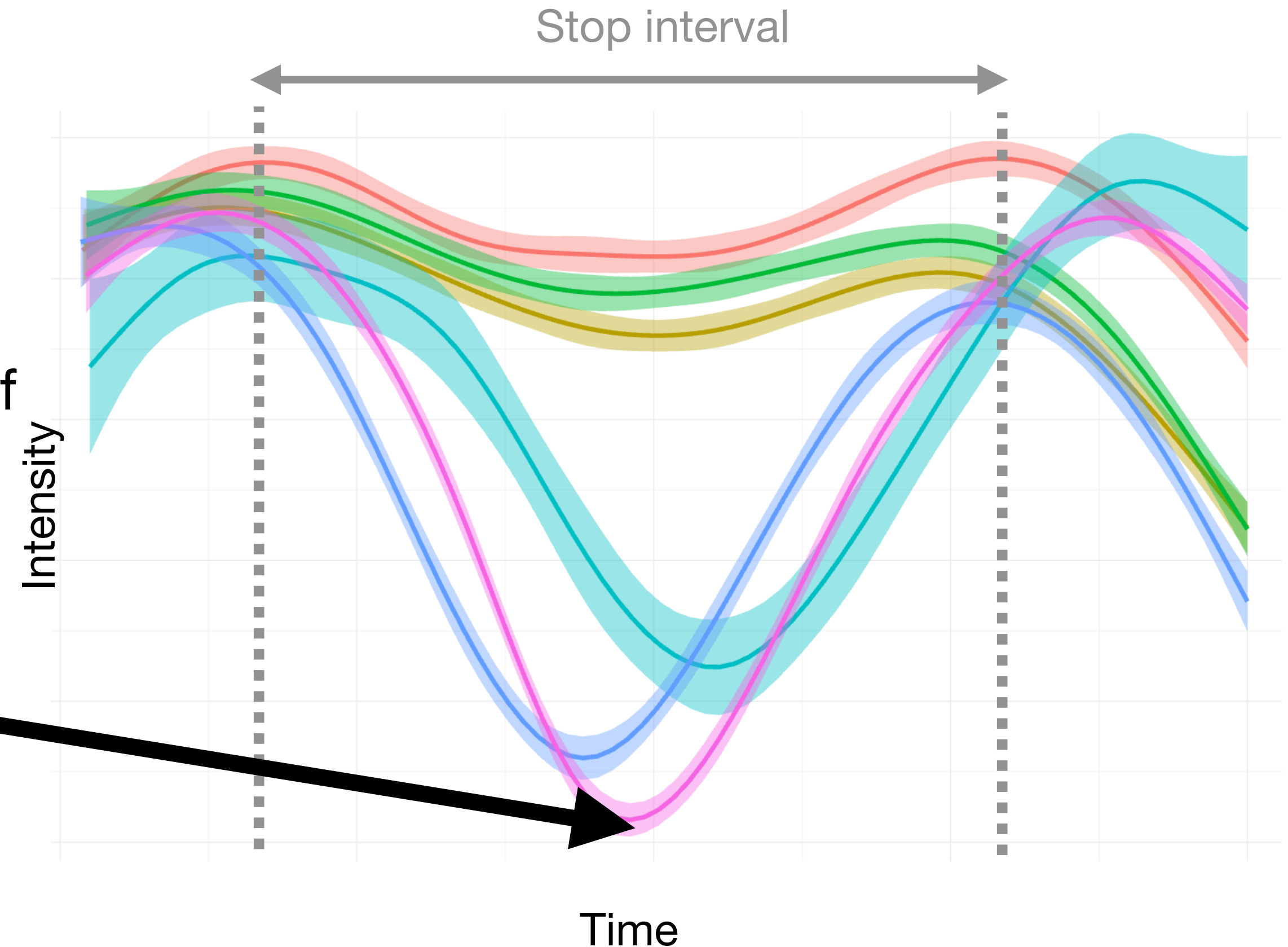
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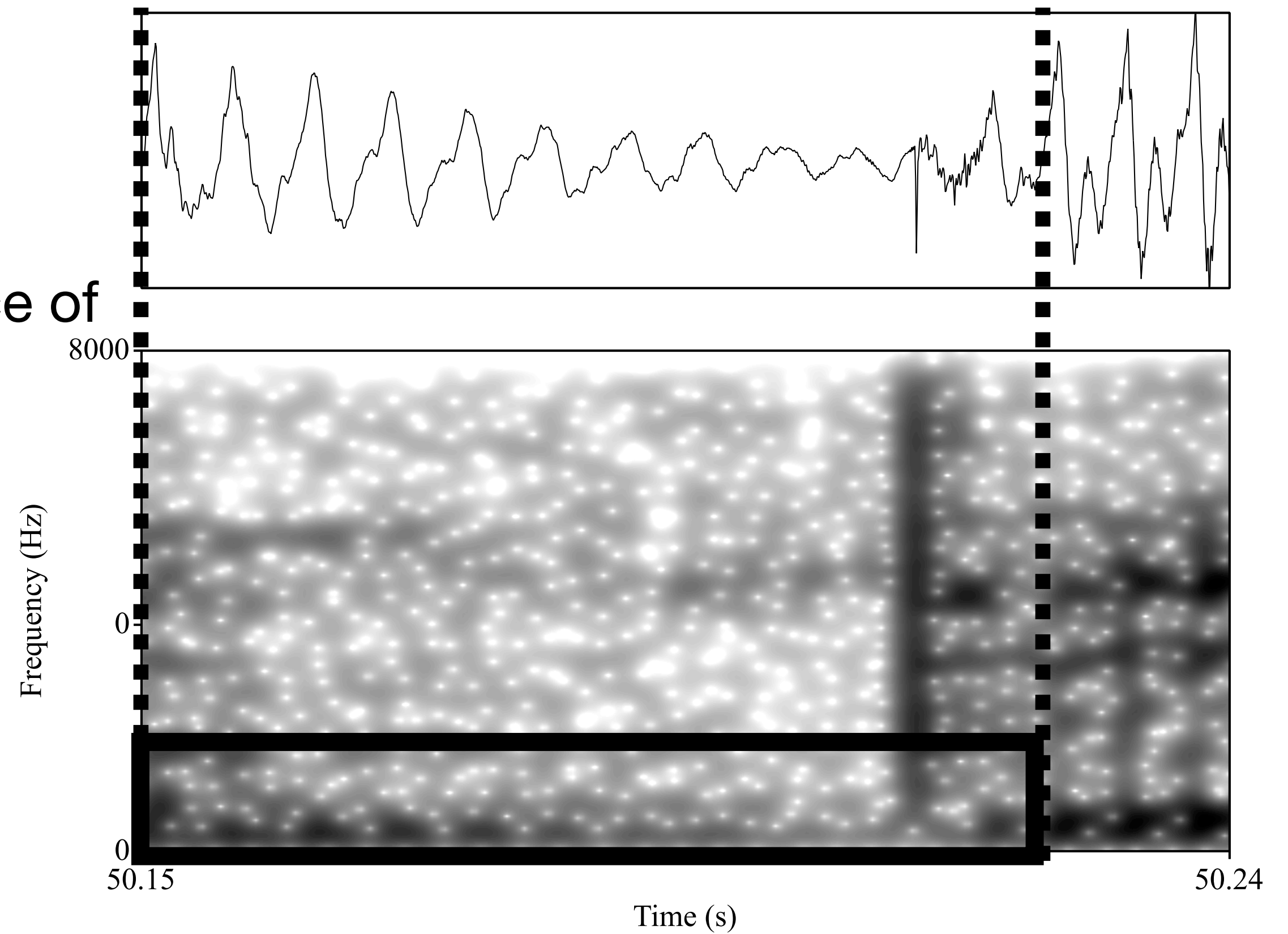
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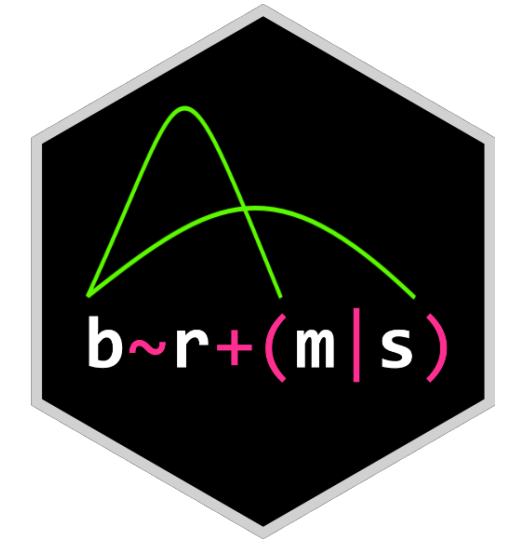


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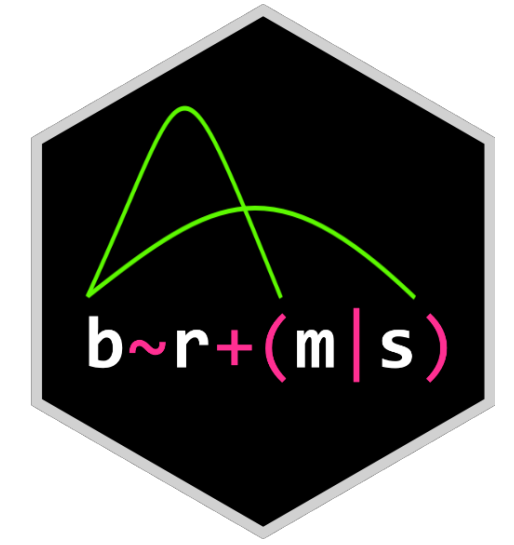


Model



- Bayesian multivariate regression model
 - Jointly-model all 4 response variables (burst presence, duration, intensity velocity, VDI)
- Fixed effects
 - Phonological voicing {voiced, voiceless}
 - PoA {labial, velar}
 - Speech rate (log)
- Random effects
 - Dialect/corpus/speaker intercepts
 - Voicing, POA, speech rate slopes
 - Cross-response {intercept, voicing} correlations

Model

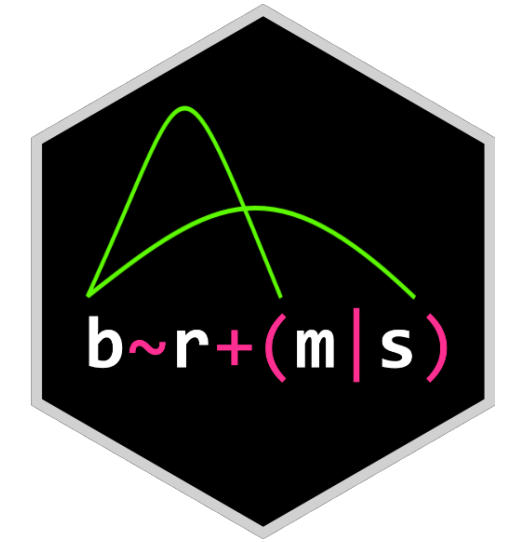


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RQ1: *how consistent is
the realisation of
phonological voicing
across English dialects?*

Allow dialects to vary in
voicing contrast size

Model

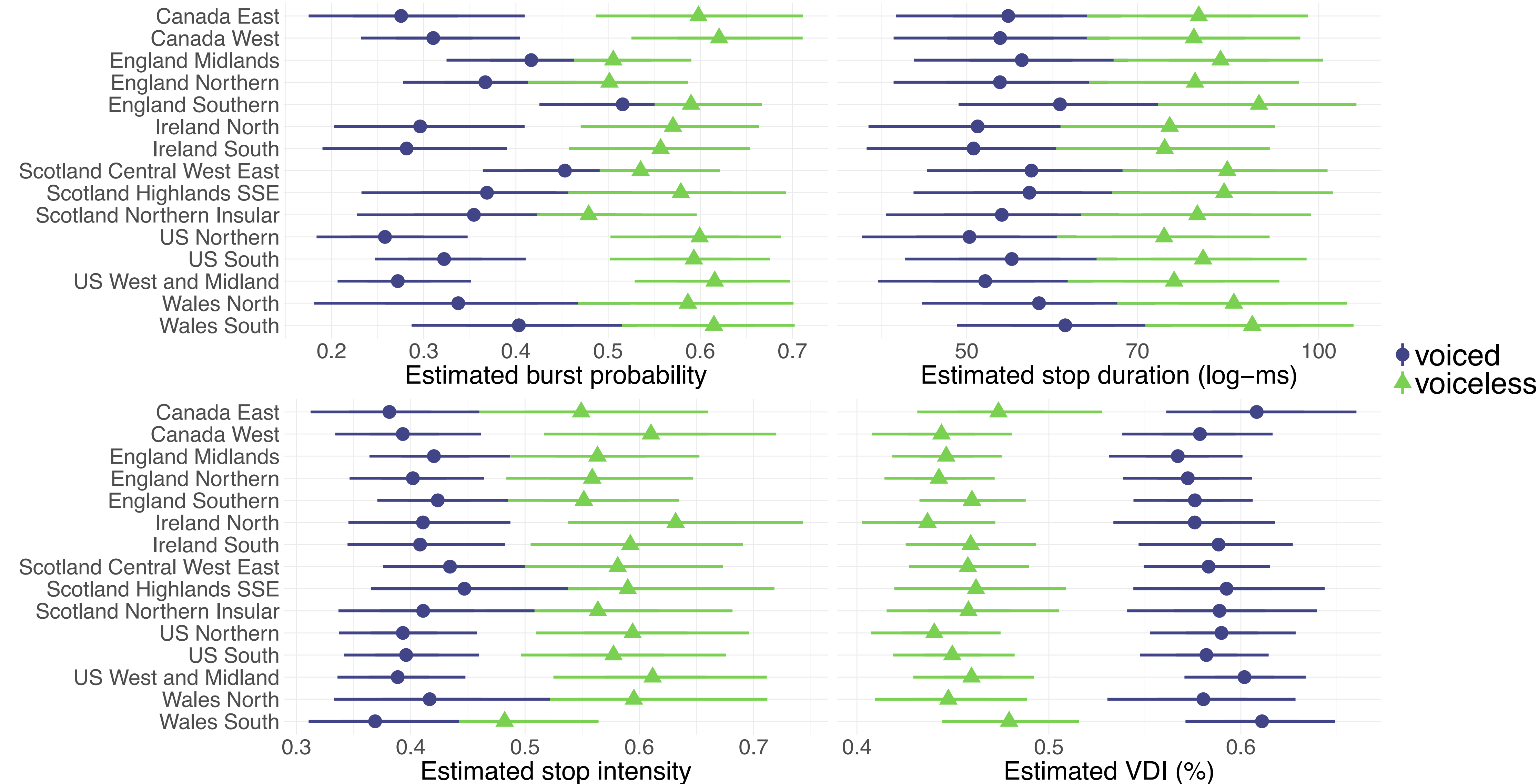


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 - **Cross-response {intercept, voicing} correlations**

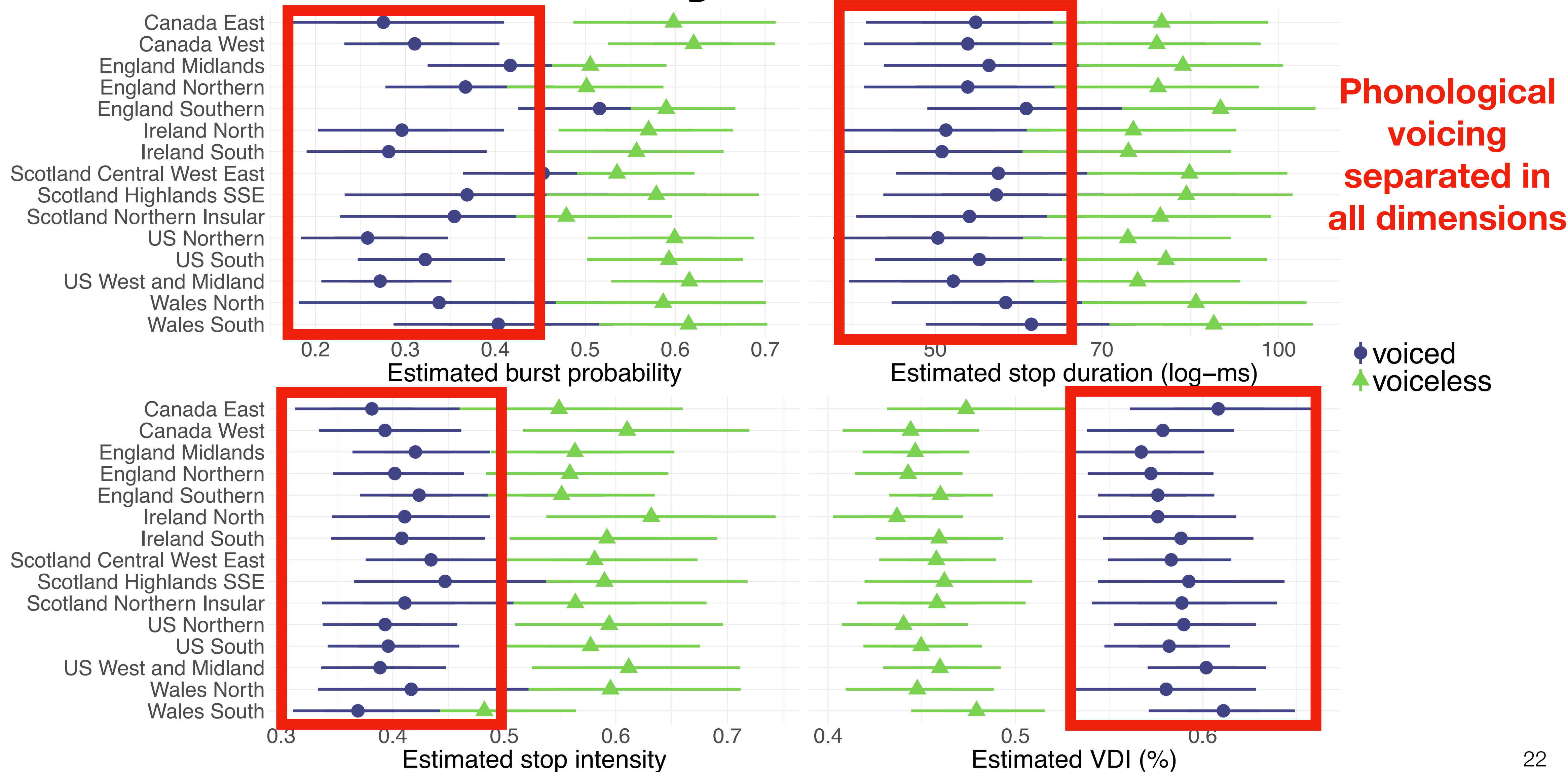
RQ2: *how are multiple acoustic properties related to each other in signalling the contrast across dialects?*

Directly model relationship between voicing contrast sizes of different responses

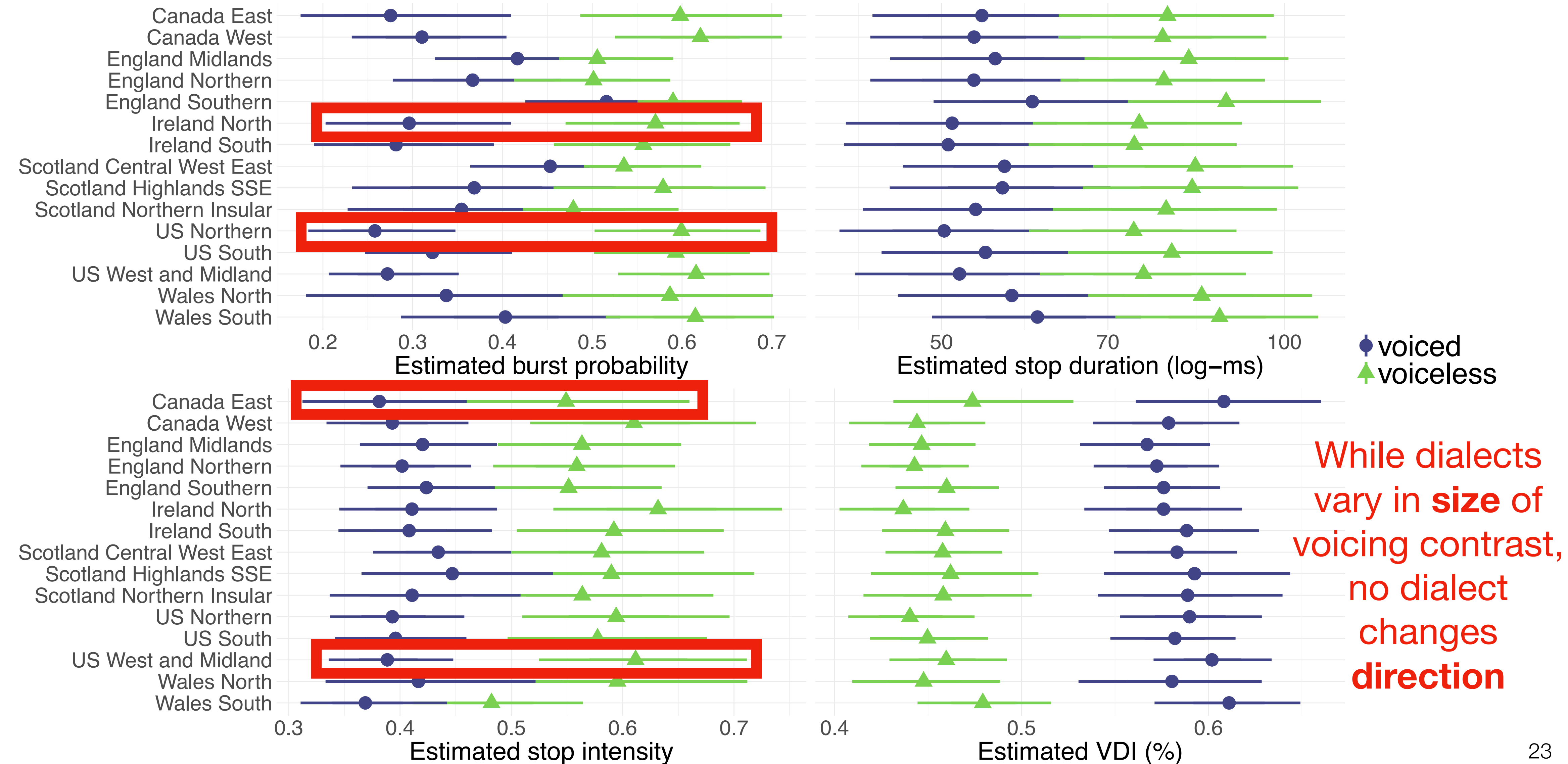
Q1: consistency across dialects



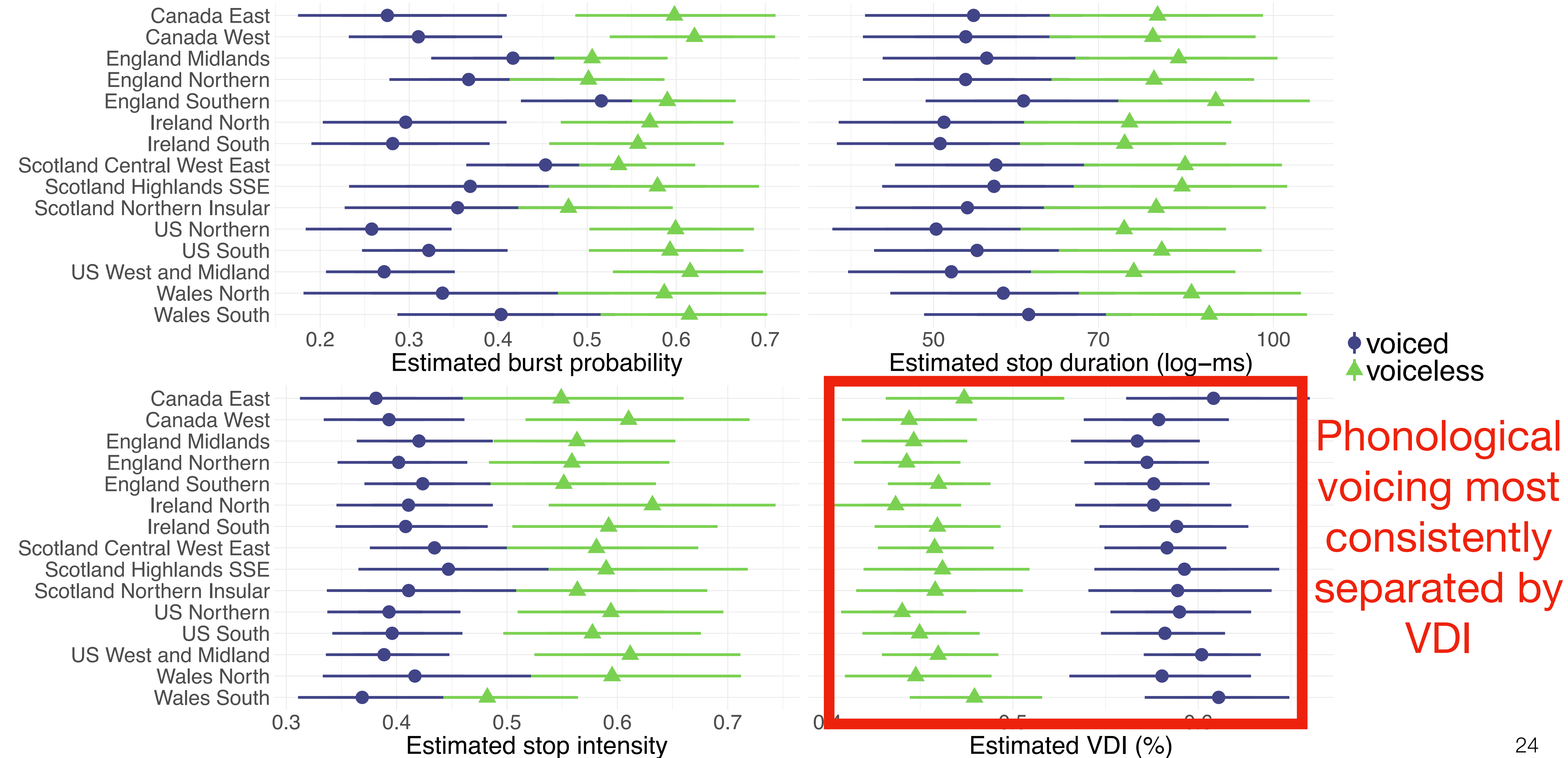
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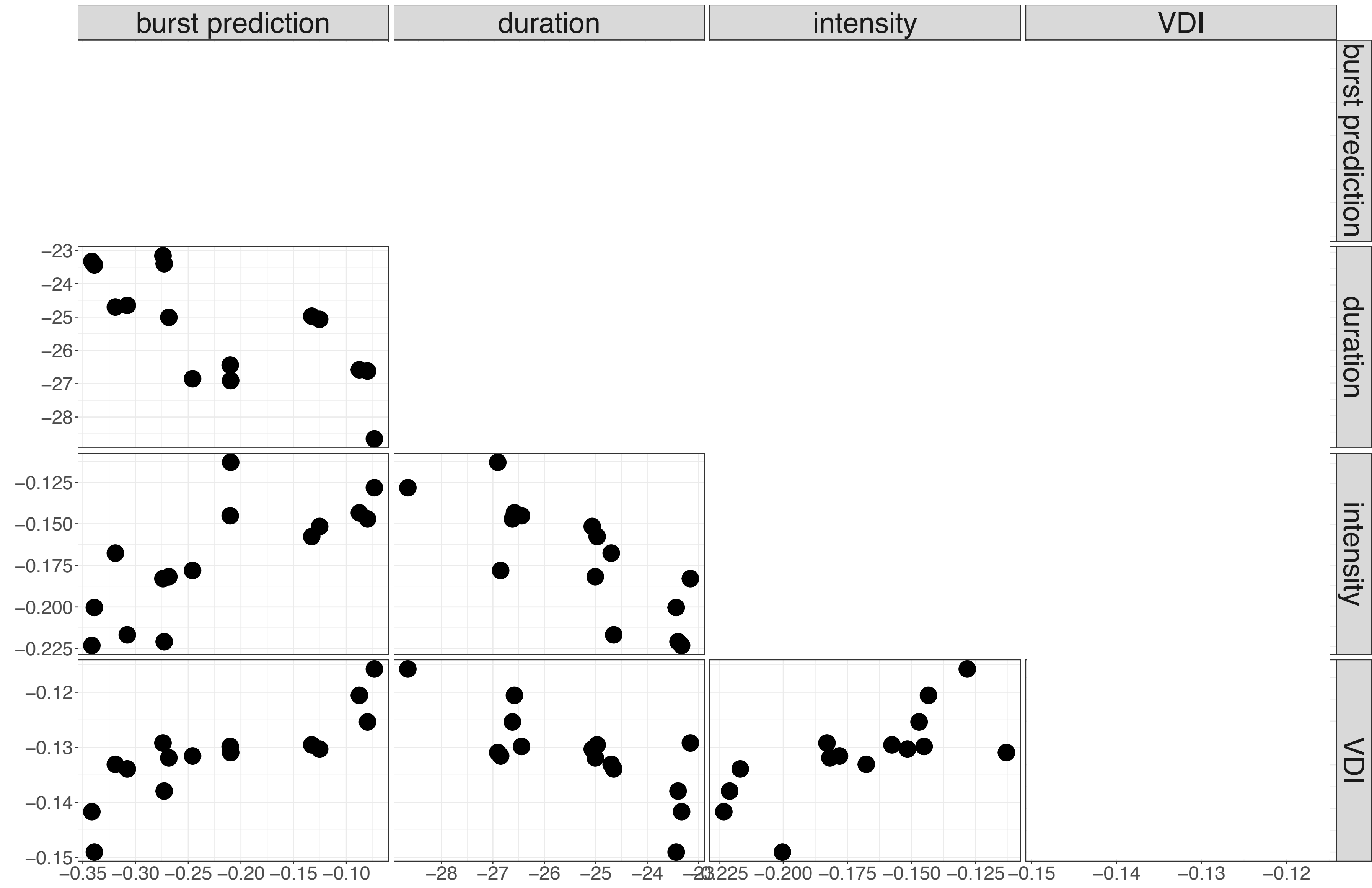
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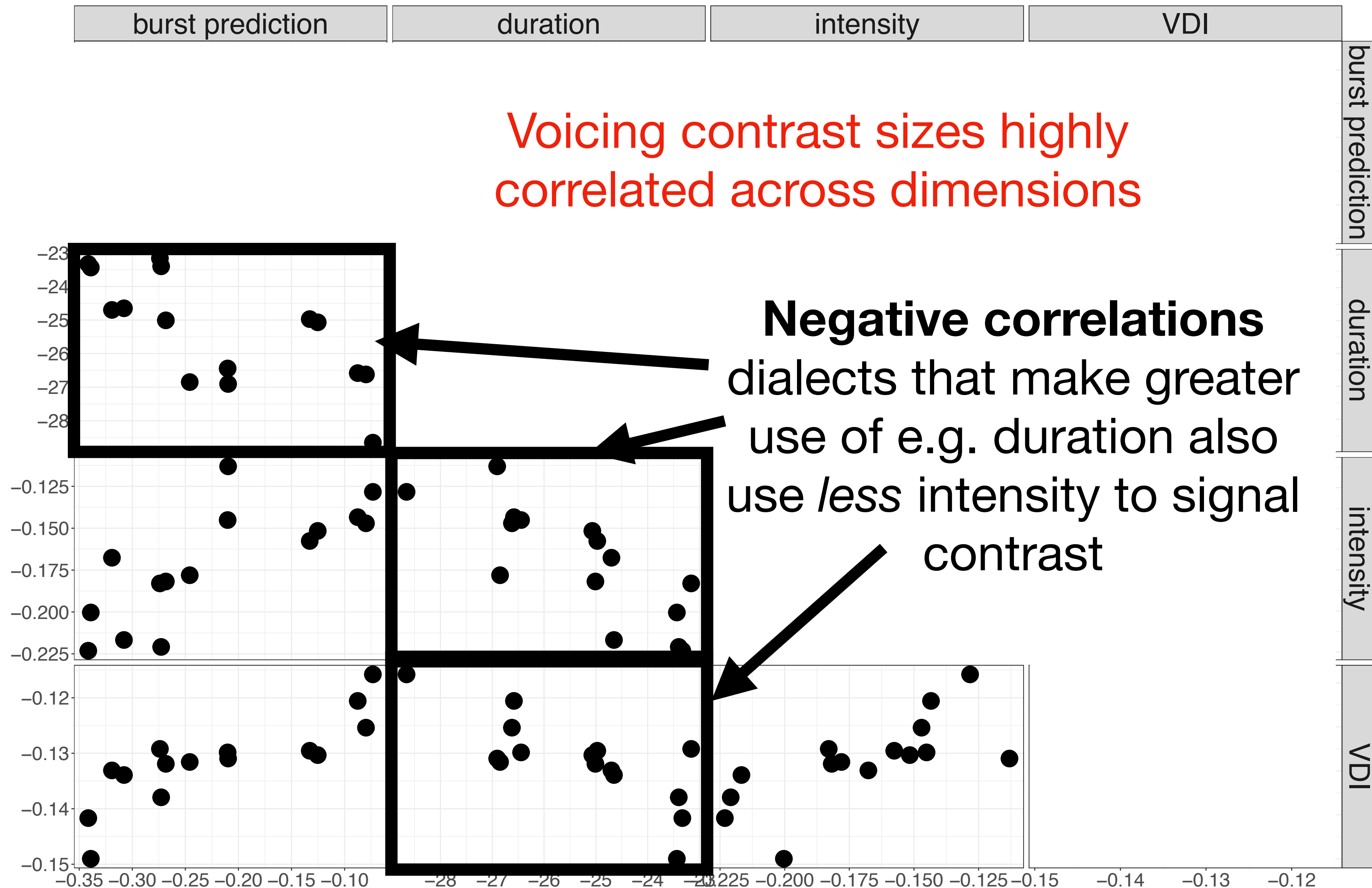
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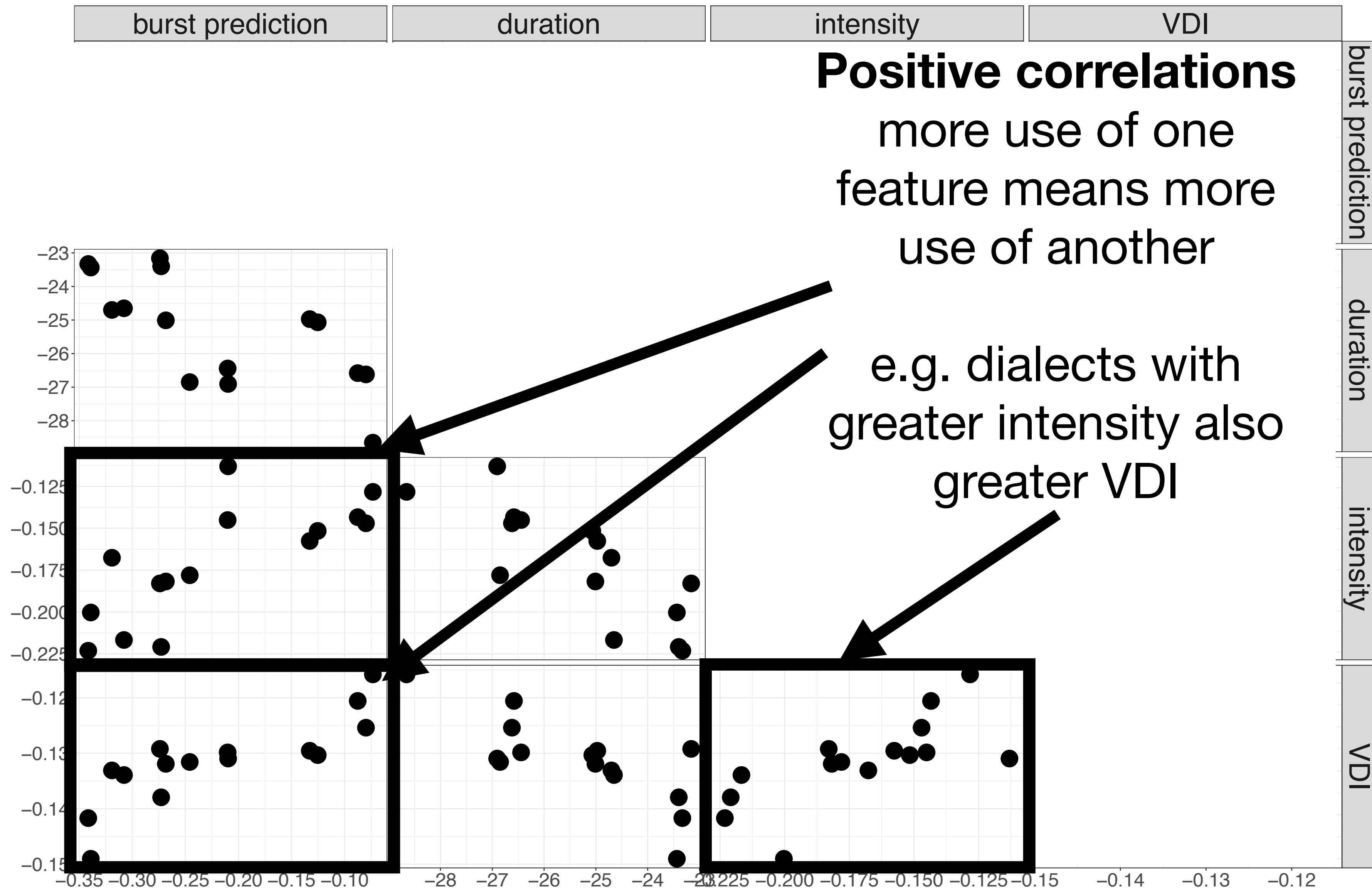
Q2: correlations across dialects



Q2: correlations across dialects



Q2: correlations across dialects



Discussion

- RQ1: *How consistent is the implementation of phonological voicing across many dialects of English?*
 - Dialects pattern similarly in burst presence, duration, intensity, and VDI in marking the phonological voicing contrast
 - Dialects vary in contrast **size** but not **direction**

Discussion

- RQ1: *How consistent is the implementation of phonological voicing across many dialects of English?*
 - Dialects pattern similarly in burst presence, duration, intensity, and VDI in marking the phonological voicing contrast
 - Dialects vary in contrast **size** but not **direction**
- RQ2: *How are multiple acoustic properties related to each other in signalling the voicing contrast across English dialects?*
 - Acoustic dimensions to the English voicing contrast are highly correlated
 - Both **positive** and **negative** correlations across dimensions

Future work

- *To what extent do features map onto a single spectrum of how voicing is realised?*
 - Explore low-dimensional space of variability (via dimensionality reduction)
- *How much cross-feature structure is shared across individual speakers?*
 - *Do speakers vary more than dialects?*
 - *Are speakers correlated in the same way as dialects?*

Thank you!



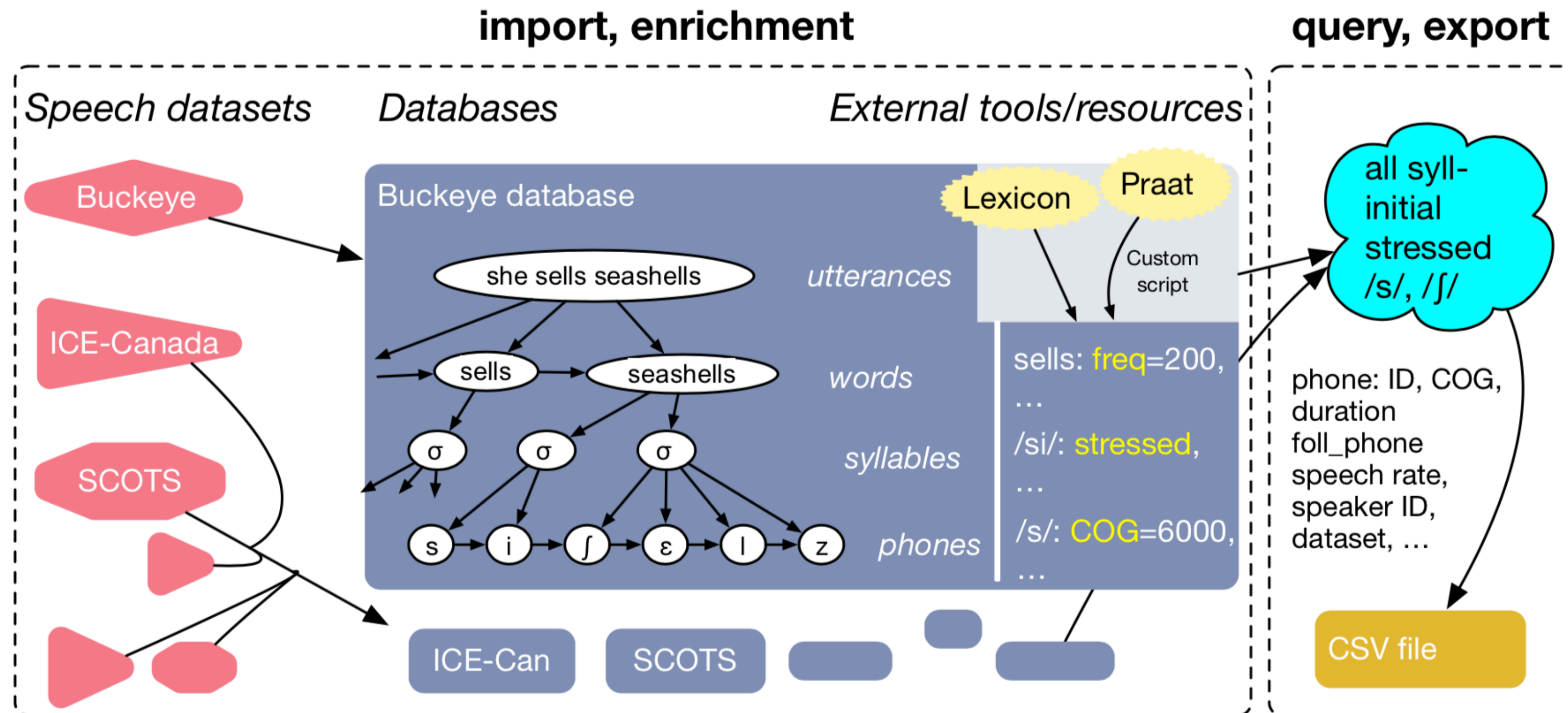
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PolyglotDB extraction



Dataset

Dialect	speakers	tokens
Canada_East	11	532
Canada_West	332	16782
England_Midlands	920	6943
England_Northern	697	5400
England_Southern	1861	21359
Ireland_North	66	1021
Ireland_South	206	1102
Scotland_Central_West_East	355	13728
Scotland_Highlands_SSE	8	218
Scotland_Northern_Insular	9	274
US_Northern	77	2291
US_South	288	22231
US_West_and_Midland	465	30168
Wales_North	33	66
Wales_South	80	1015

Measures: burst presence

Interspeech 2025
17-21 August 2025, Rotterdam, The Netherlands



Automatic classification of stop realisation with wav2vec2.0

James Tanner¹, Morgan Sonderegger², Jane Stuart-Smith¹, Jeff Mielke³, Tyler Kendall⁴

¹Department of English Language and Linguistics, University of Glasgow, United Kingdom

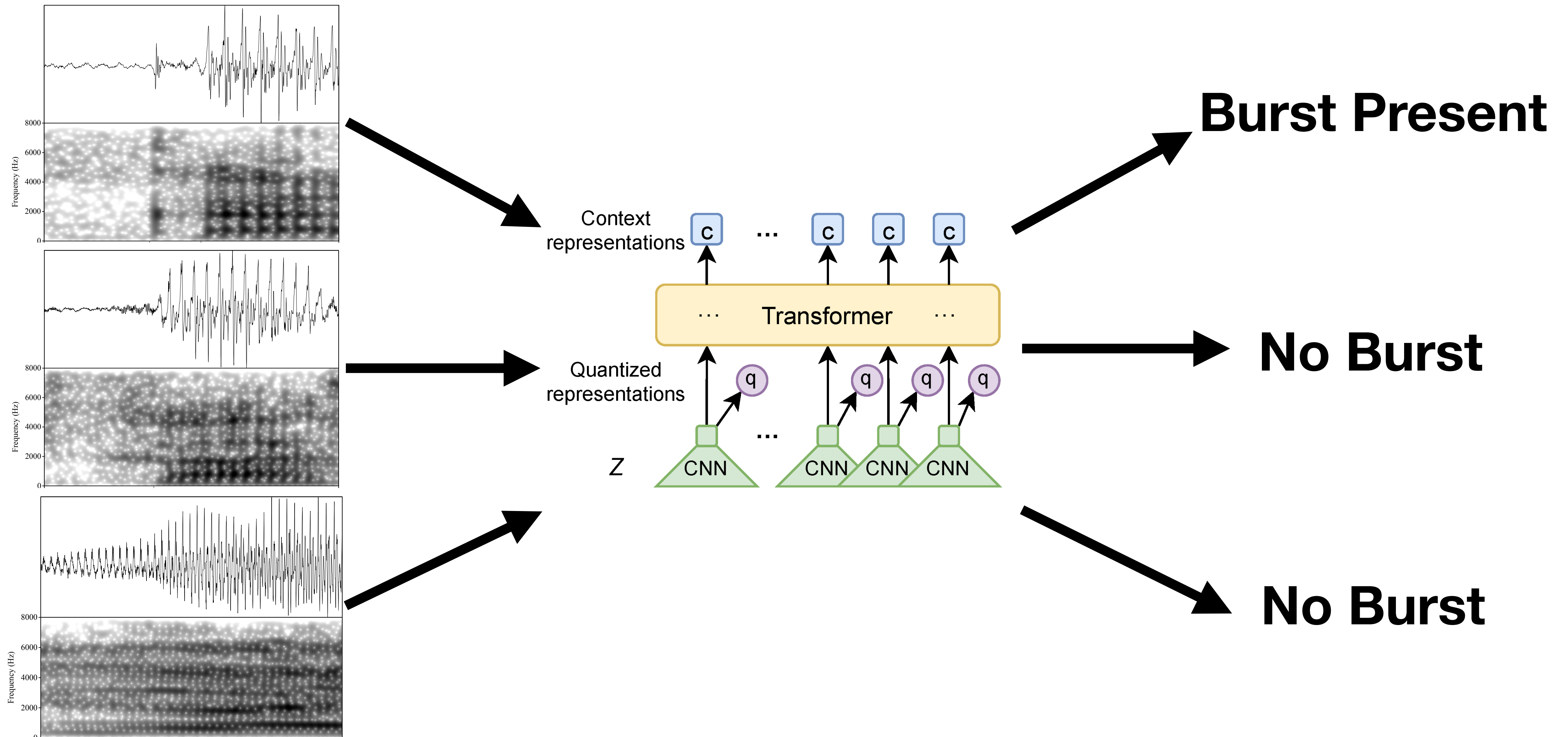
²Department of Linguistics, McGill University, Canada

³Department of English, North Carolina State University, United States

⁴School of Law, Duke University, United States

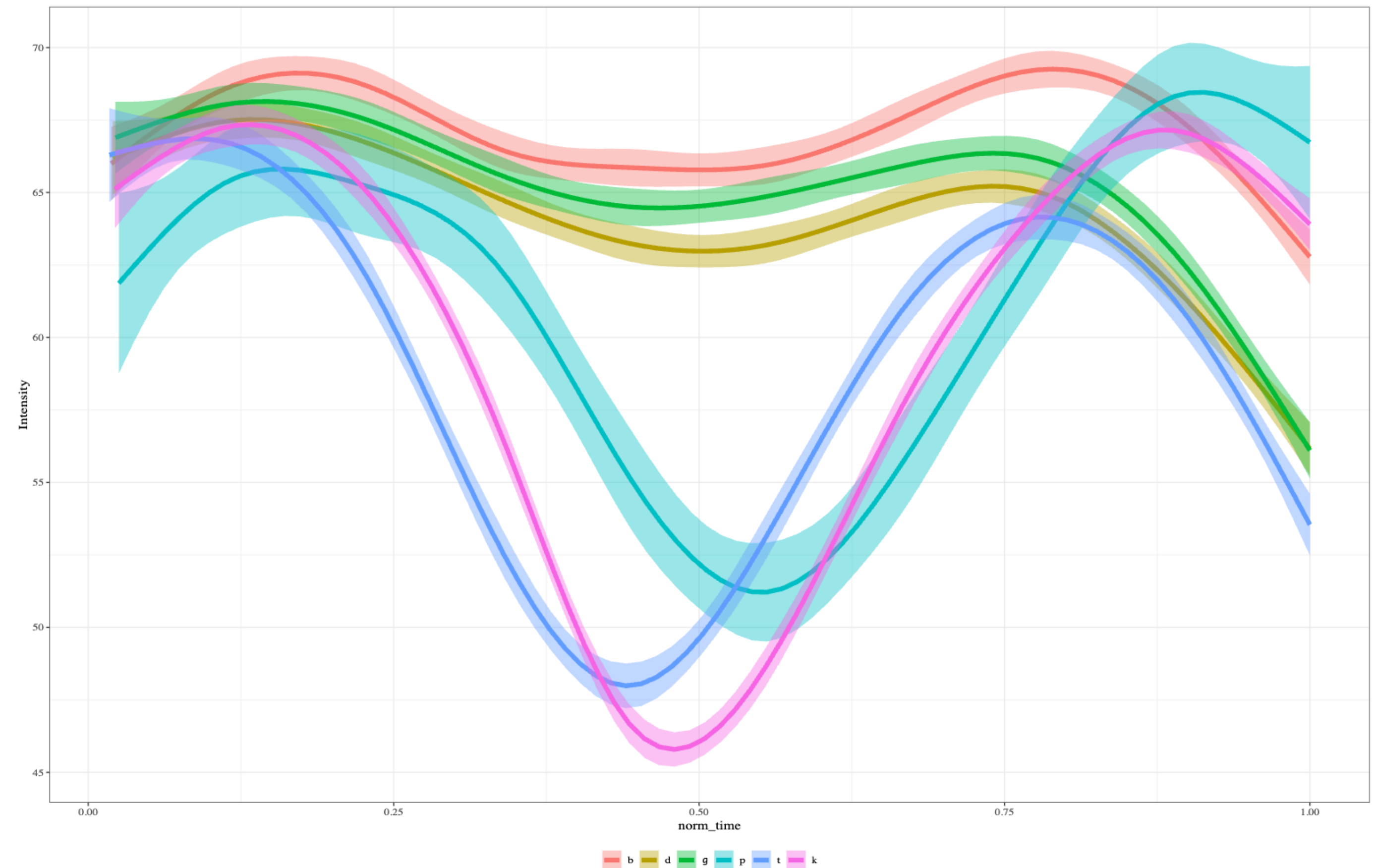
- Pre-trained wav2vec2.0 model
- Fine-tuned to predict binary (presence/absence) of stop bursts
 - ~300k Japanese stops (Corpus of Spontaneous Japanese)
 - ~50k English stops (SPADE)

Measures: burst presence



Measures: intensity

- Generate intensity track (min pitch = 100Hz, time step = 1ms)
- Fit smoothing spline of VCV window and take minimum

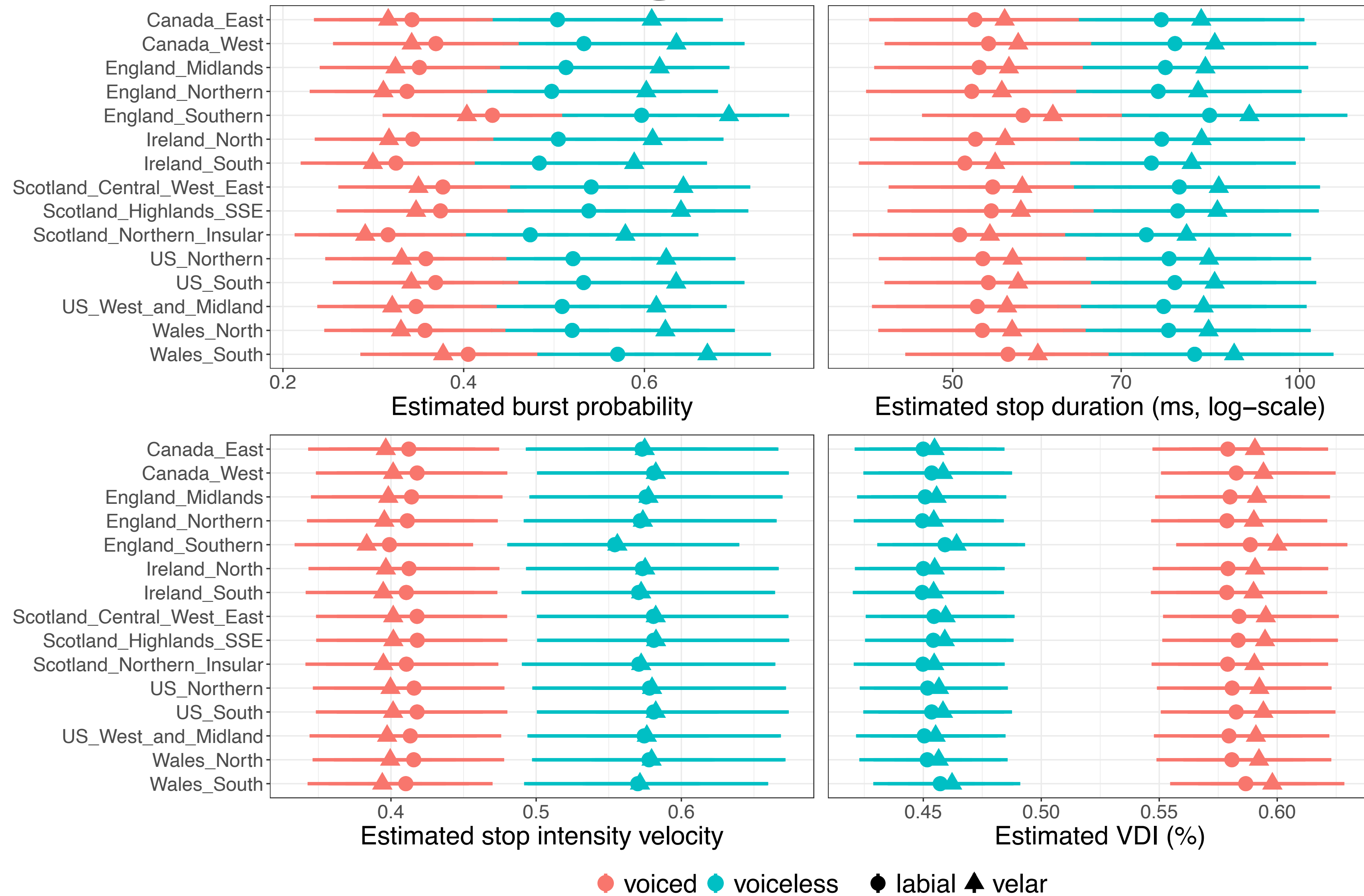


Model Structures

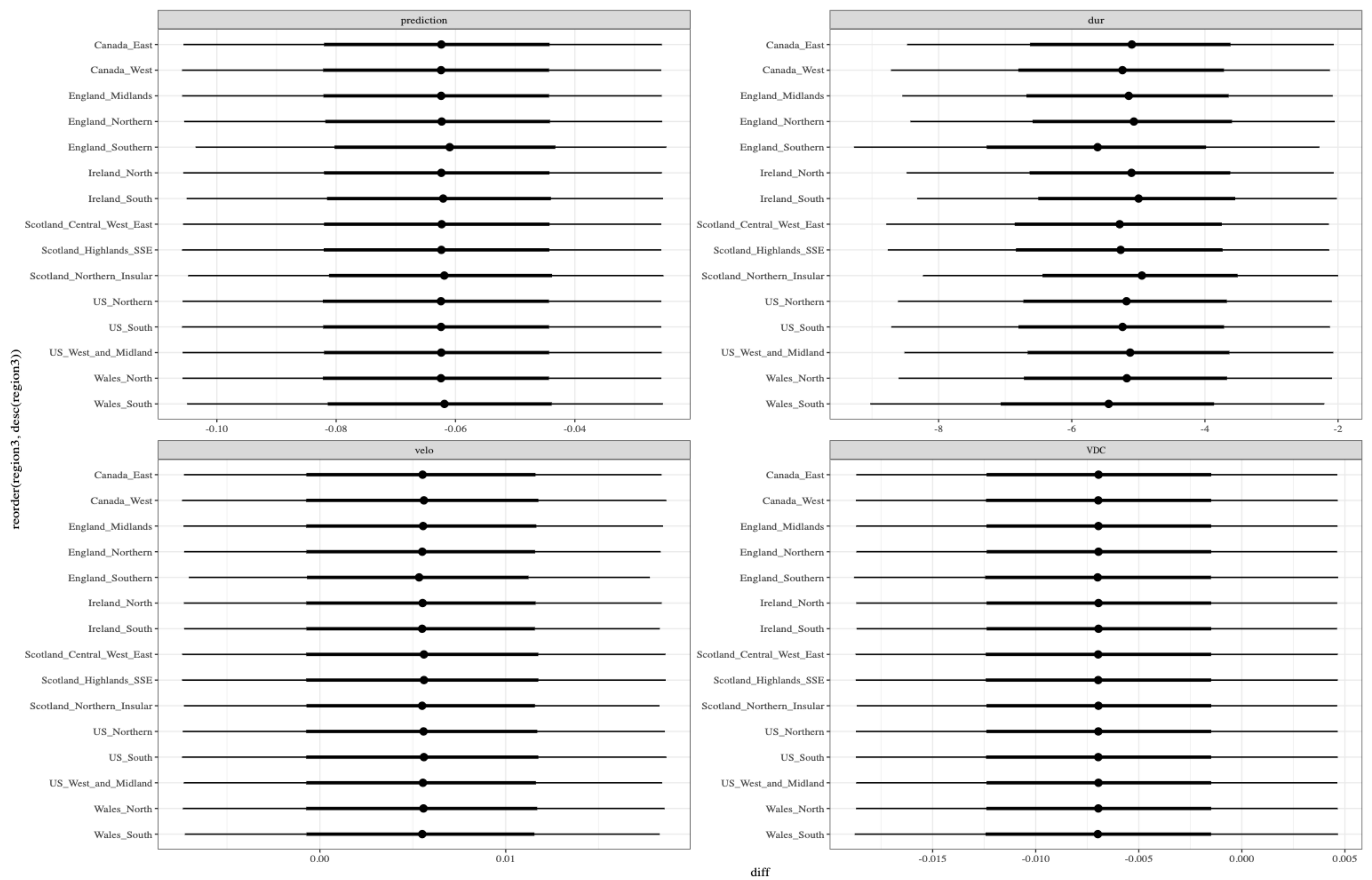
- Response families
 - Burst presence: binomial/logistic; Duration: log-normal; Intensity velocity: log-normal; VDI: zero-one-inflated beta
- Formula

response ~ voicing + PoA + ns(speech rate, 5) + voicing : PoA +
(1 + voicing + PoA + voicing : PoA | dialect) +
(0 + ns(speech rate, 5) || dialect) +
(1 + voicing + PoA + voicing : PoA | corpus) +
(0 + ns(speech rate, 5) || corpus) +
(1 + voicing + PoA + voicing : PoA | speaker) +
(1 | word)
- Priors
 - Intercepts
 - Burst = Normal(0,0.8); Dur = Normal(0,2); Intensity = Normal(0,2); VDI = Normal(0,2)
 - Betas = Normal(0,1)
 - Sigmas = Exponential(1)
 - Correlations = LKJ(1.5)

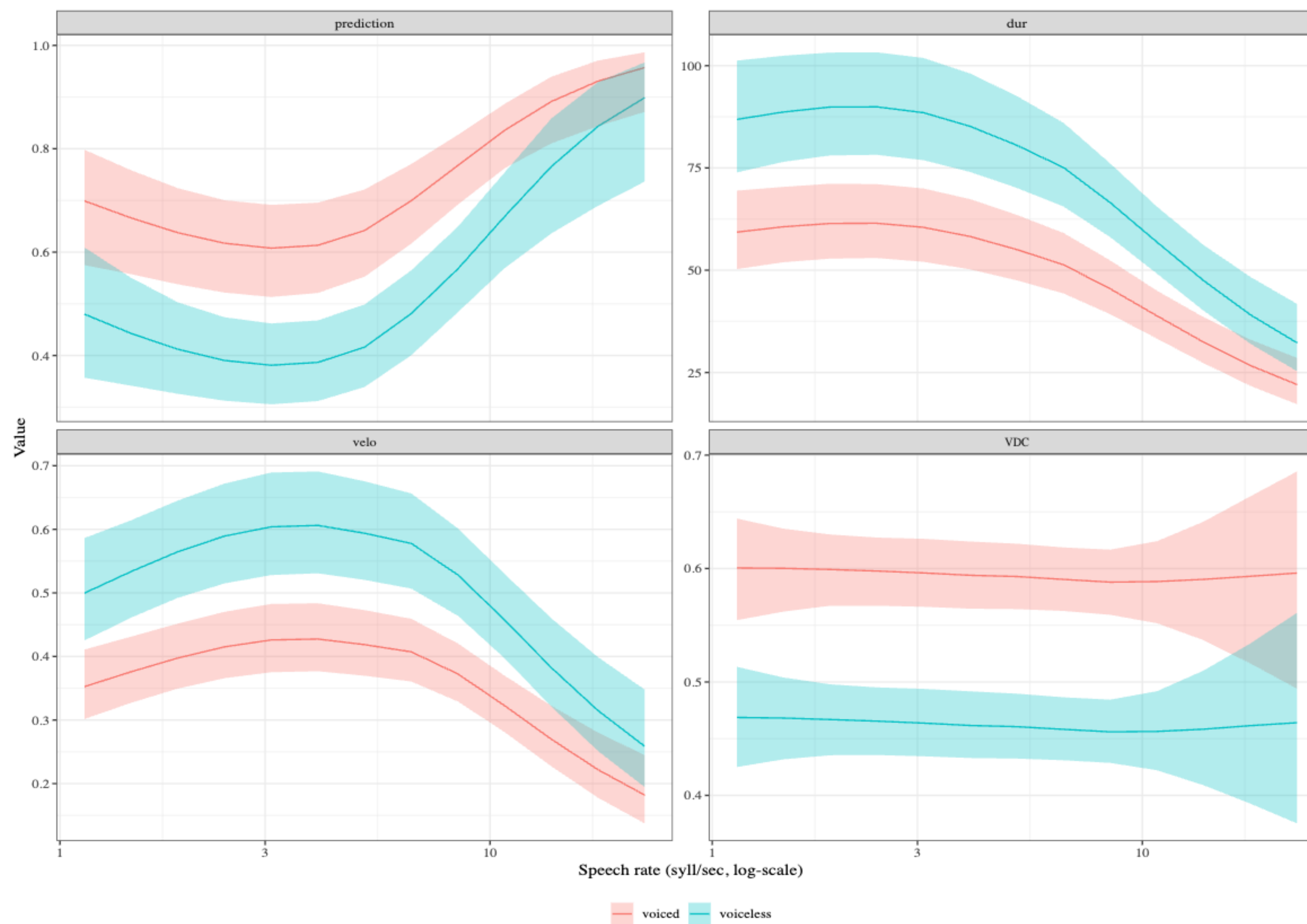
Voicing & POA



Voicing & POA



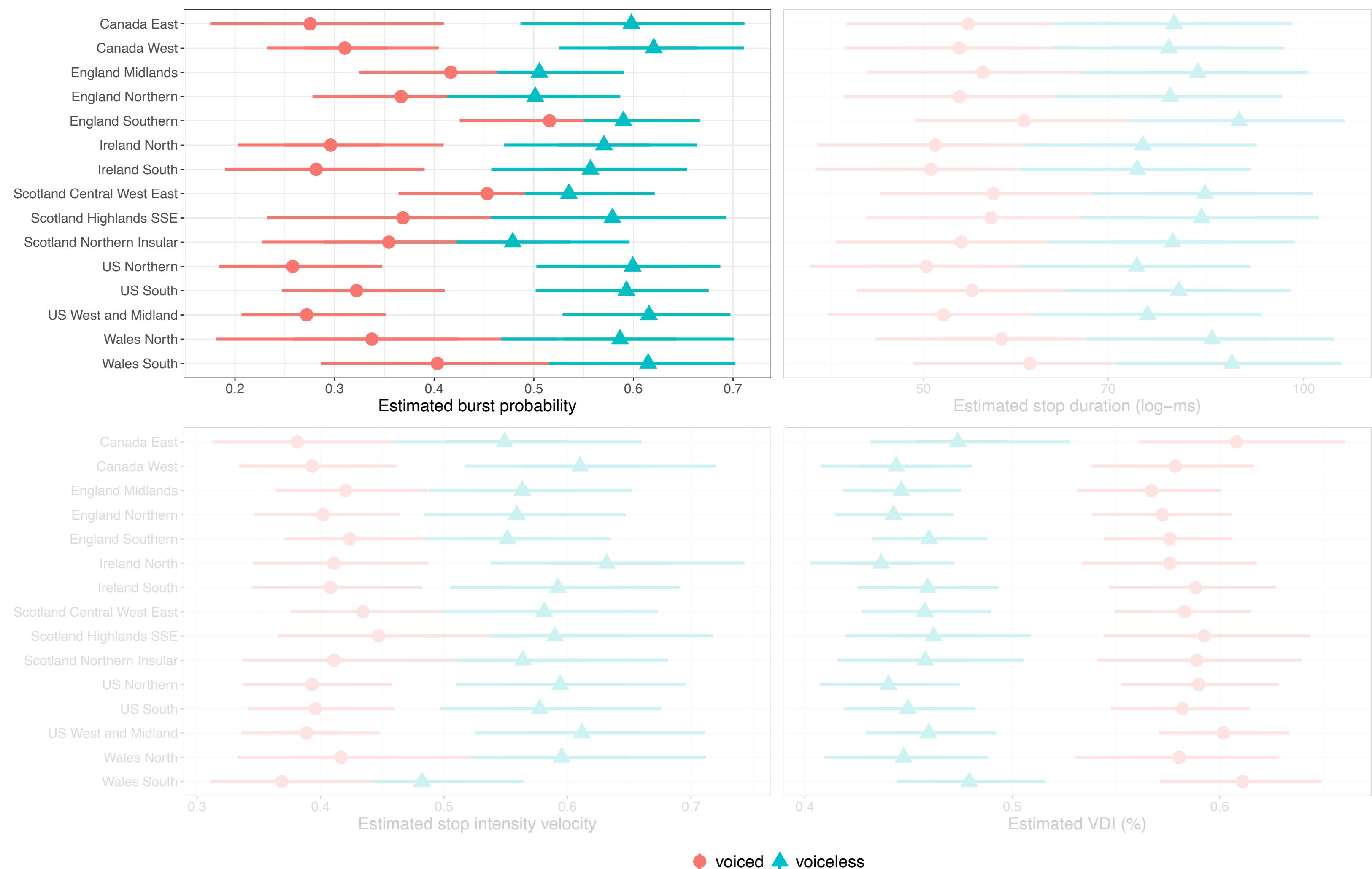
Speech rate



PCA



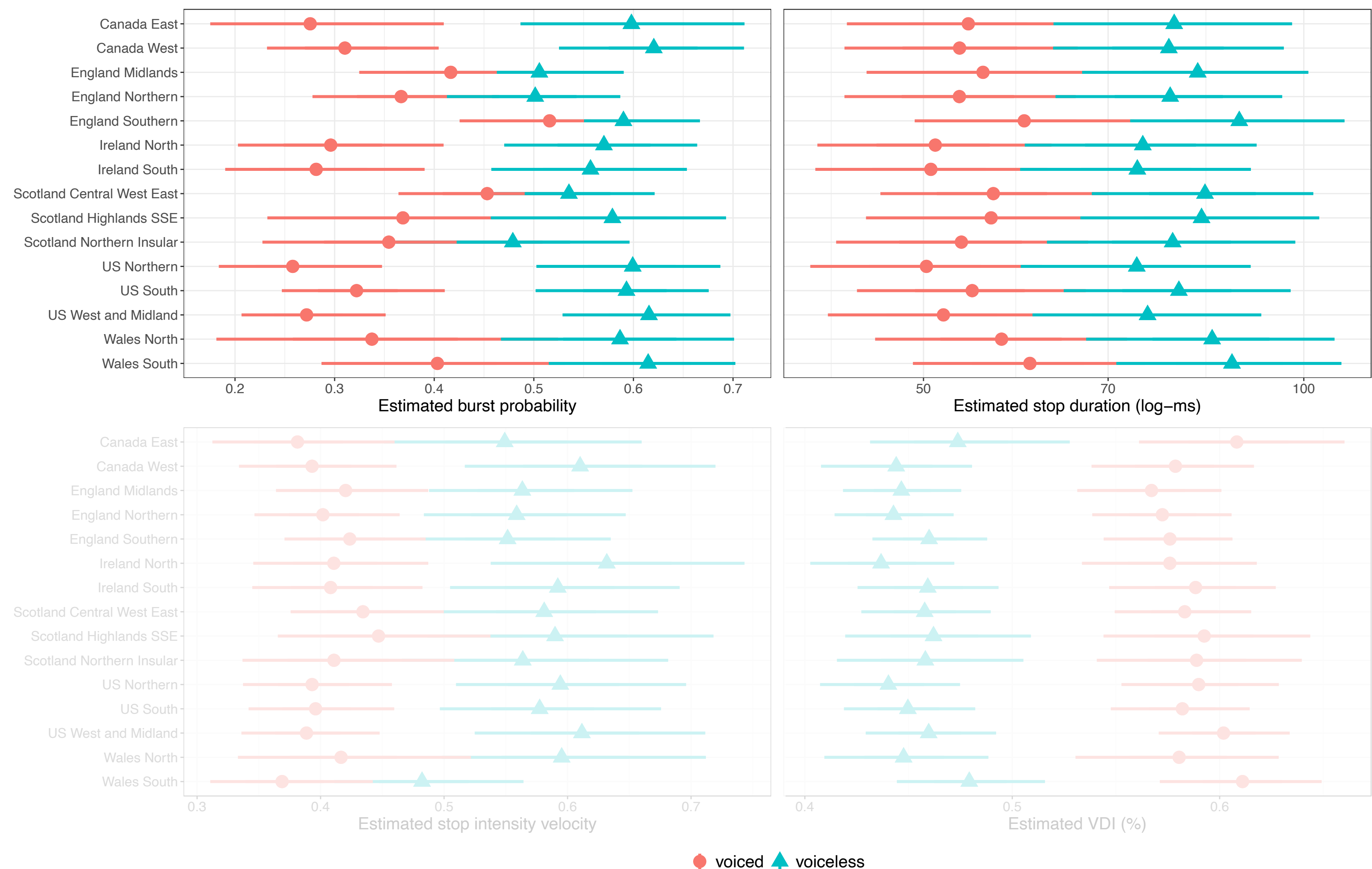
Q1: consistency across dialects



Burst presence: B = -0.9, CrI = [-1.2,-0.58]

S = 0.48, CrI = [0.31,0.77] 43

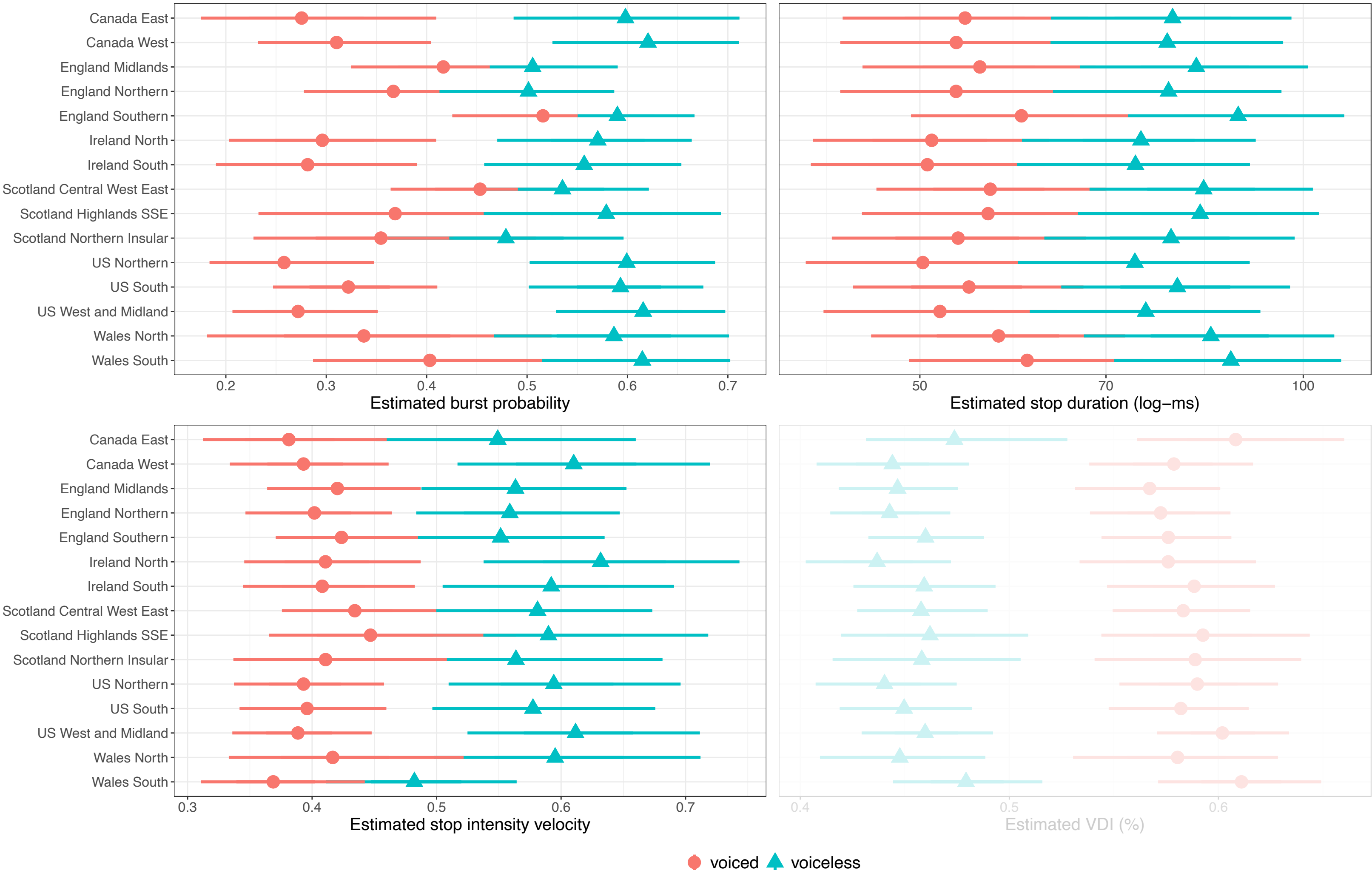
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Duration: B = -0.38, CrI = [-0.44,-0.32]

S = 0.02, CrI = [0,0.05] 44

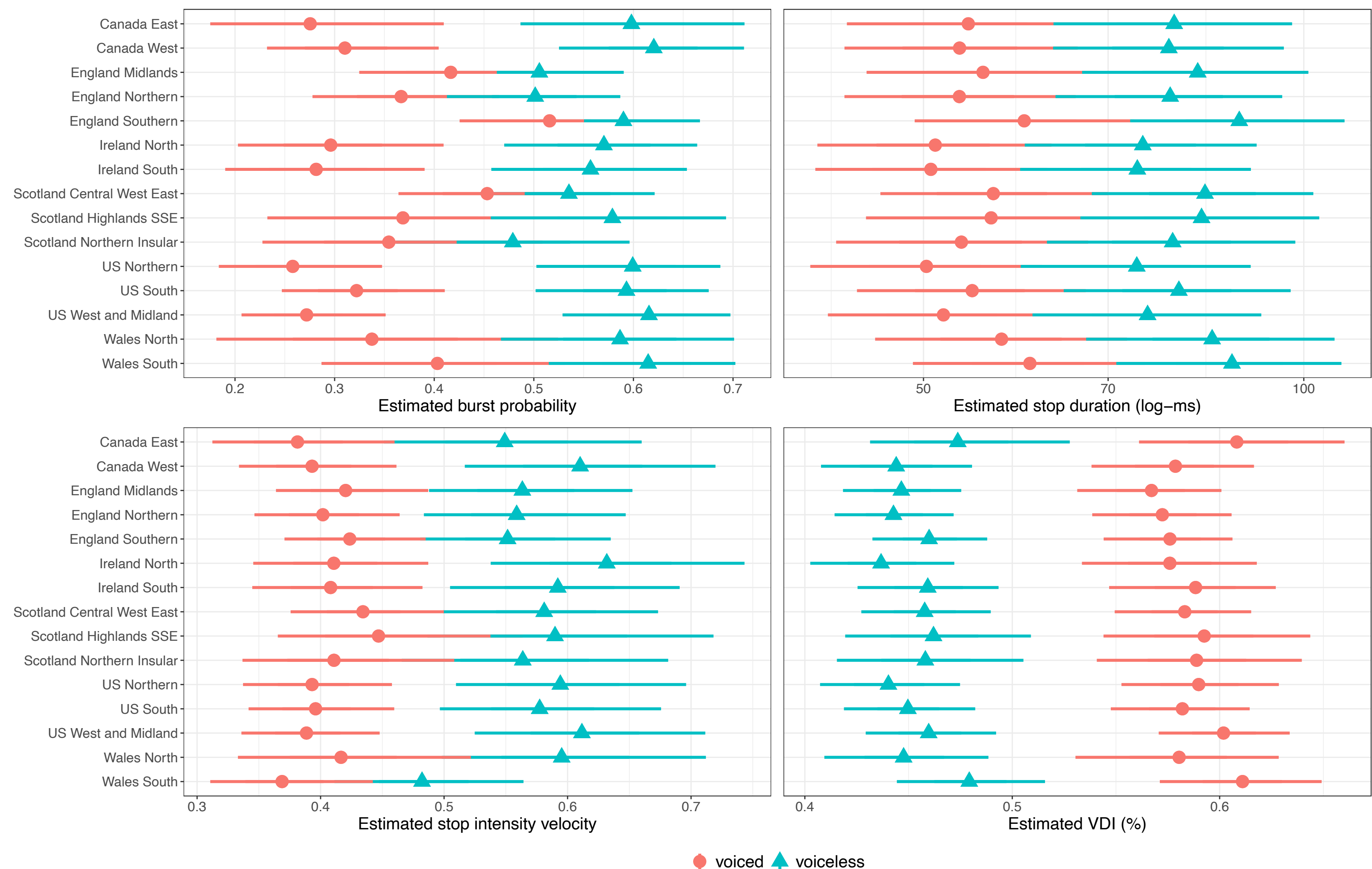
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Intensity velocity: B = -0.35, CrI = [-0.42,-0.27]

S = 0.08, CrI = [0.04,0.15] 45

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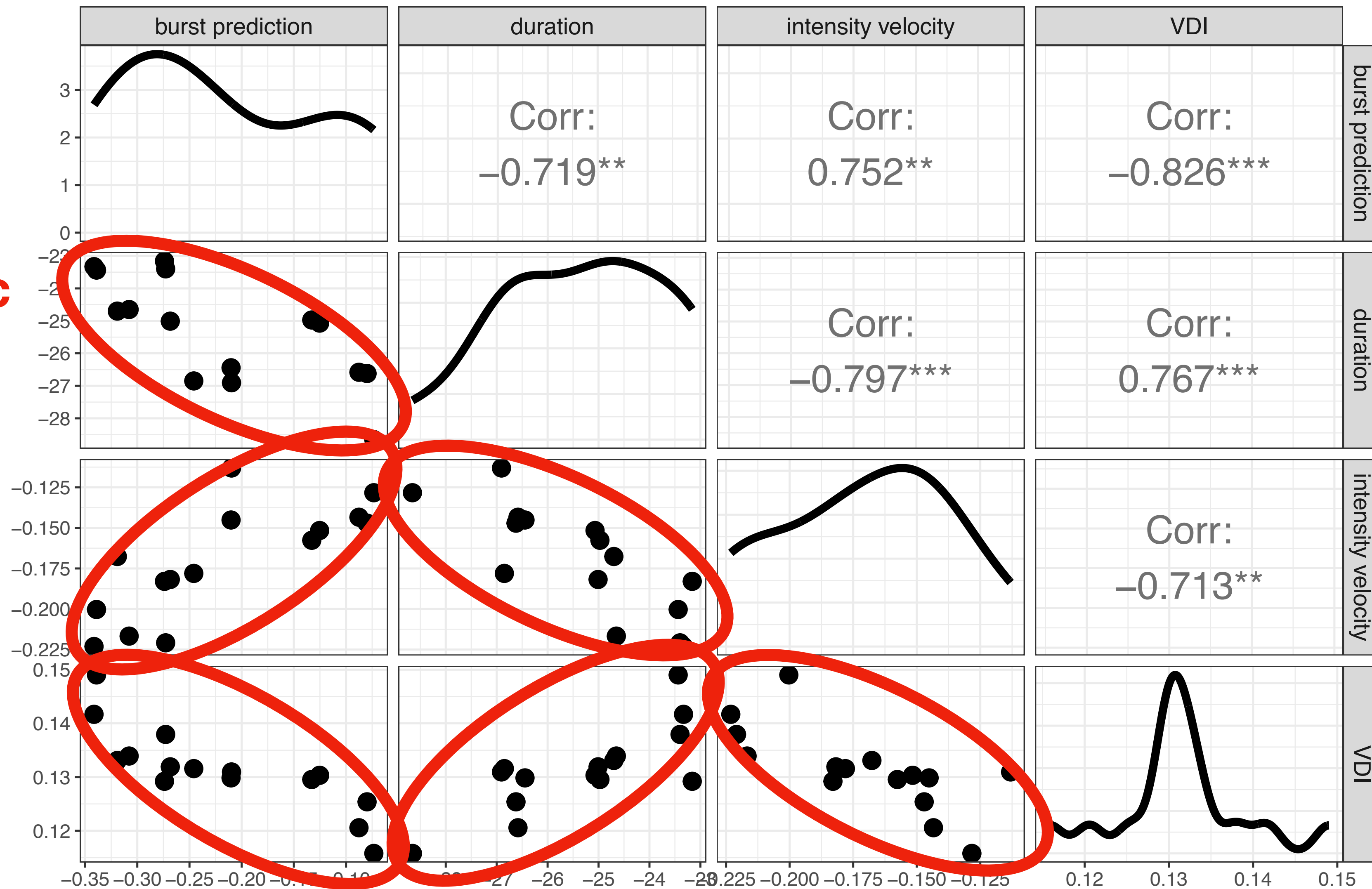
VDI: B = 0.79, CrI = [0.68,0.9]

S = 0.09, CrI = [0.02,0.18] 46

Q2: relationships across dialects

**Voicing
contrast sizes
highly
correlated
across acoustic
dimensions**

**E.g. dialects
that make
greater use of
duration also
use *less*
intensity**



Discussion

- Acoustic dimensions to phonological voicing are robust across English dialects
 - Points towards English voiced stops as consistently more lenited than voiceless
- Most acoustic dimensions are **negatively** correlated across dialects
 - While dialects can vary (to an extent) in a given dimension, this also results in a relative change in another dimension